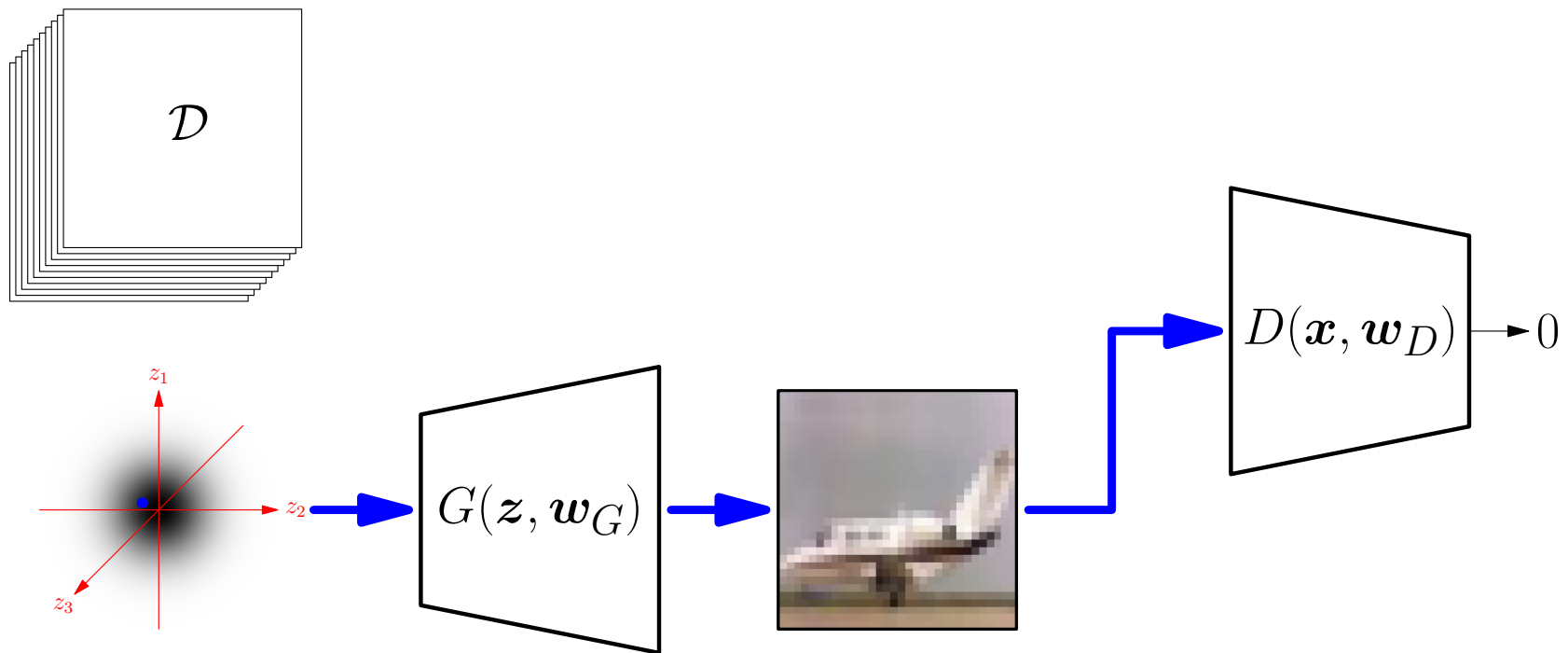


Mathematics of Machine Learning

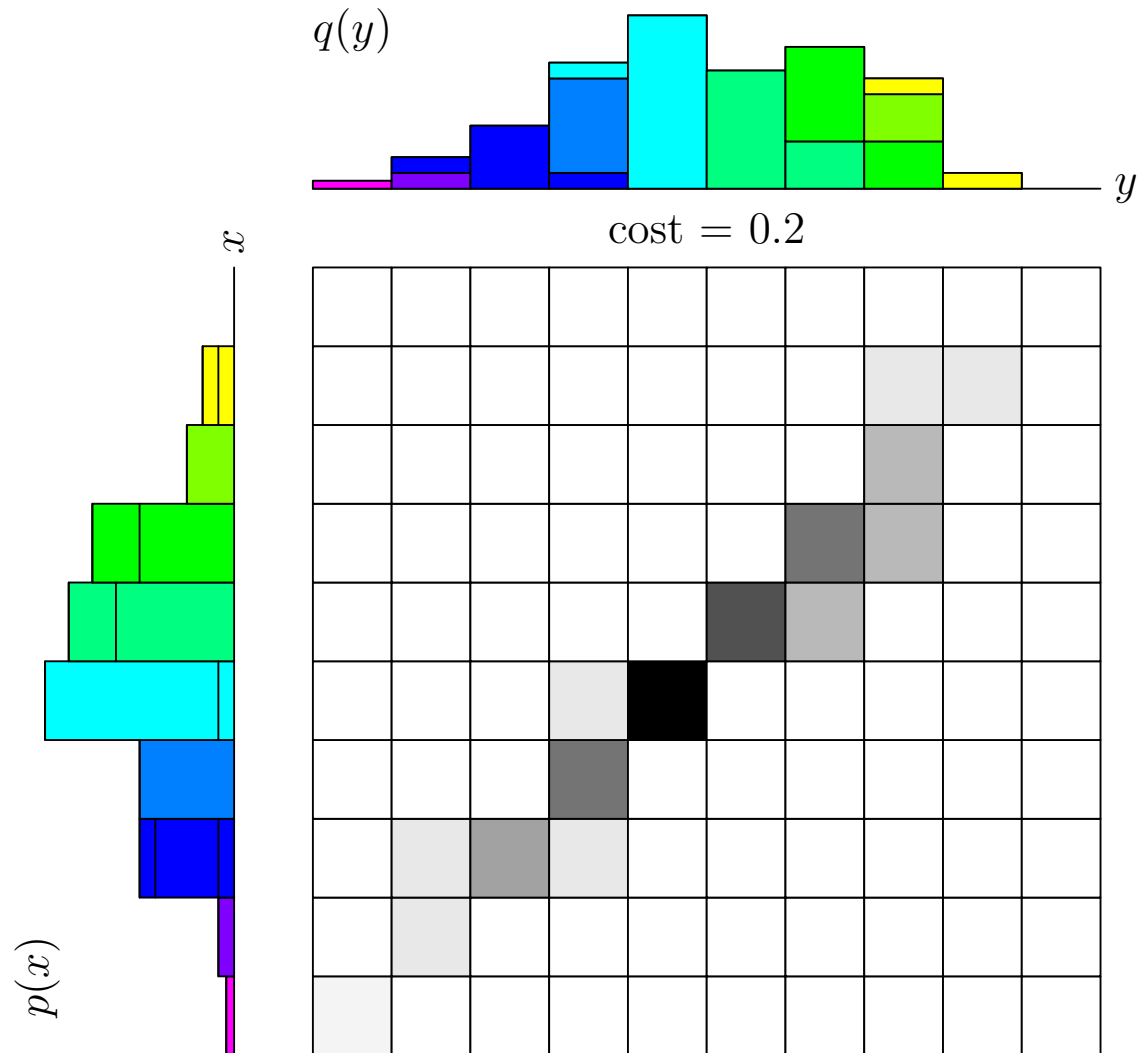
Wasserstein GANs



GANs, Wasserstein distance, Duality, WGANs

Outline

1. **GANs**
2. Wasserstein Distance
3. Wasserstein GANs



Generative Adversarial Networks

- One of the applications of Deep Learning that has most excited the public are **Generative Adversarial Networks** or GANs
- Their aim is to generate new random samples from the same distribution as some training set, $\mathcal{D}$
- Their number of real world applications are

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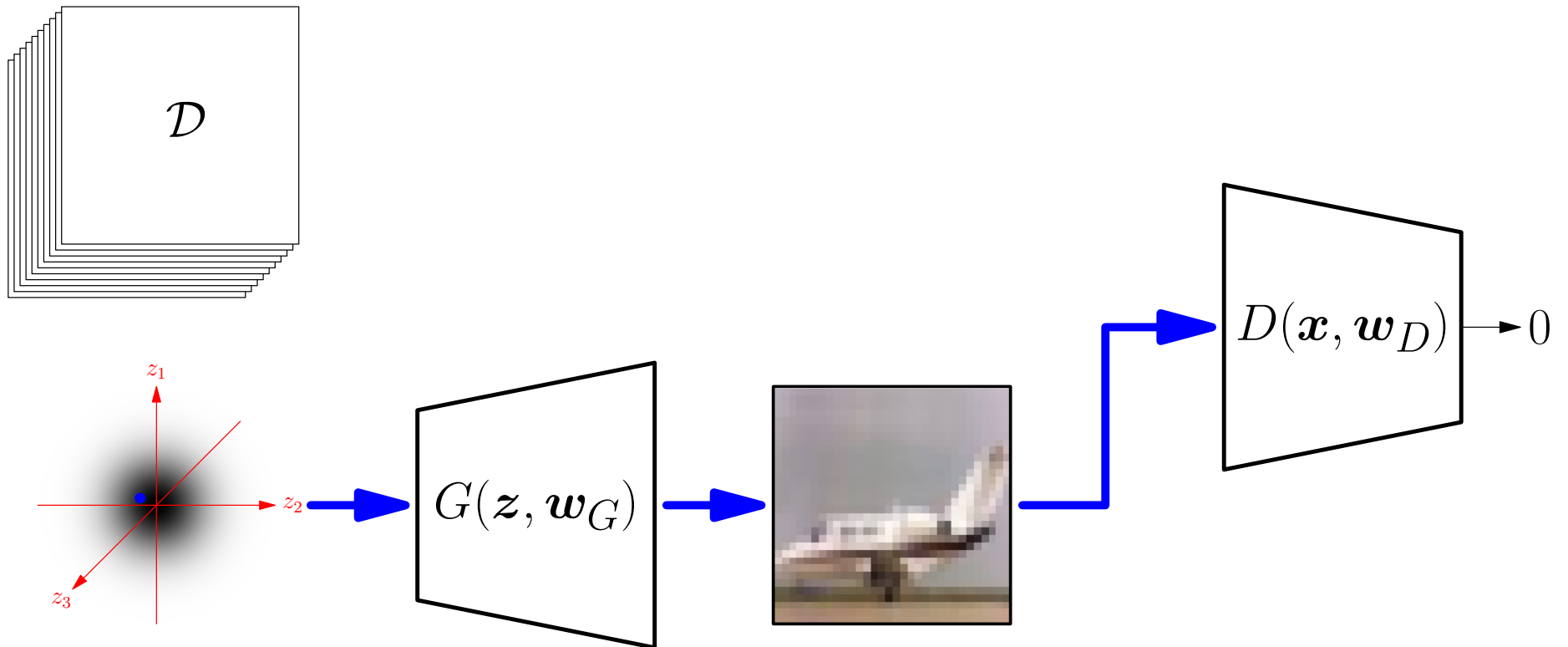
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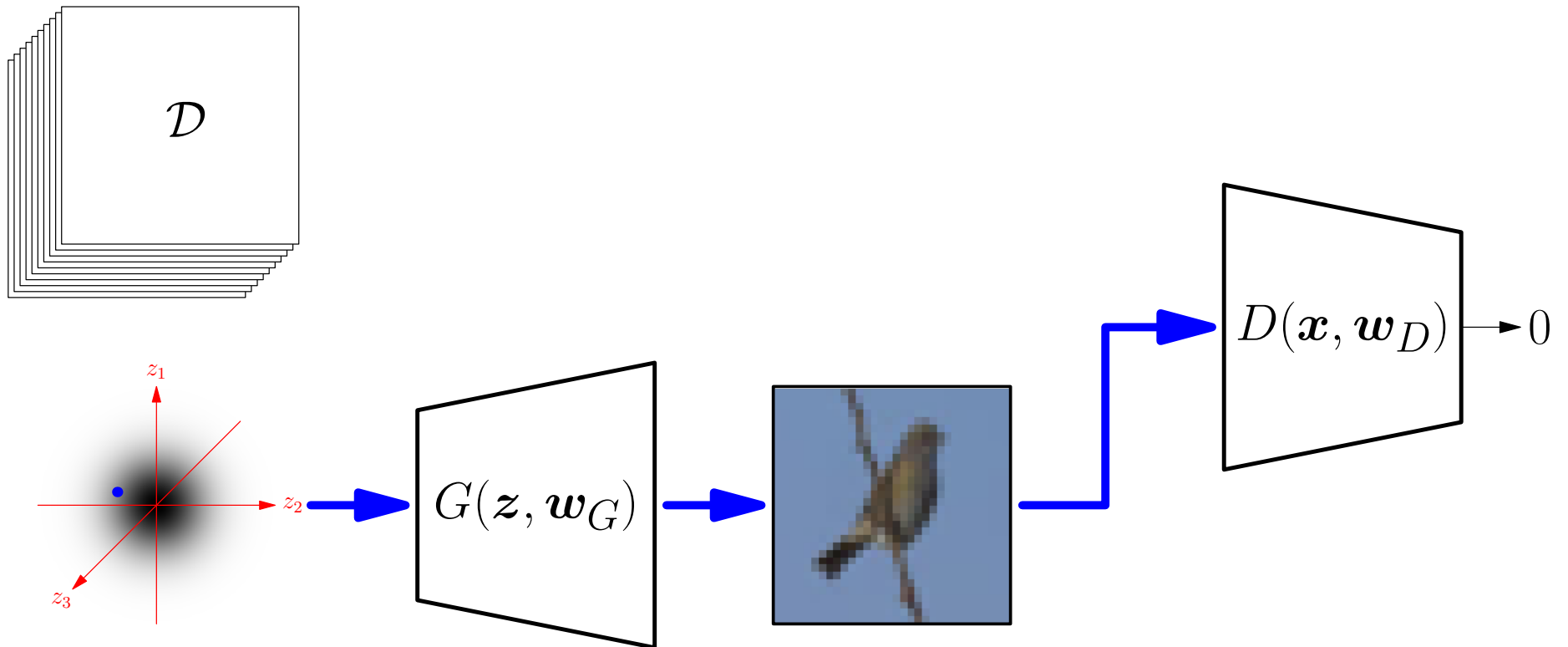
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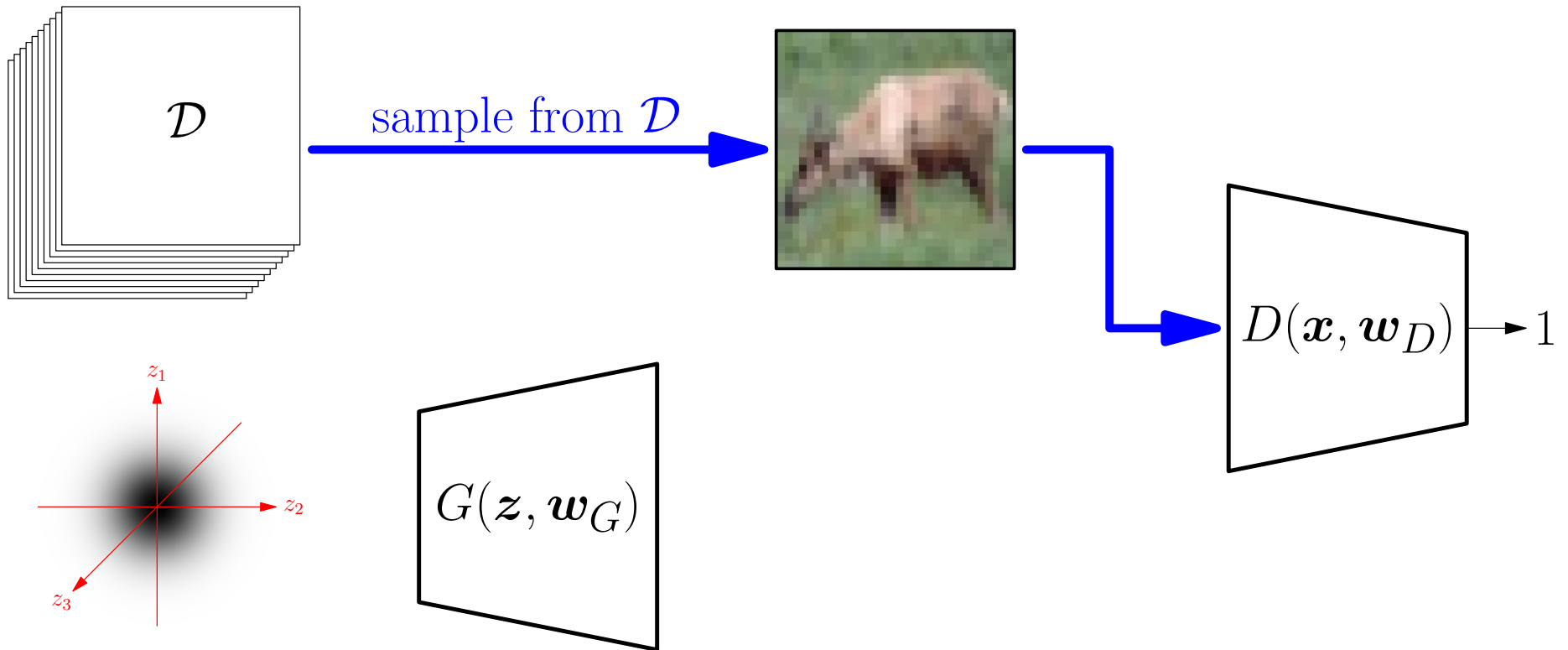
How GANs Work



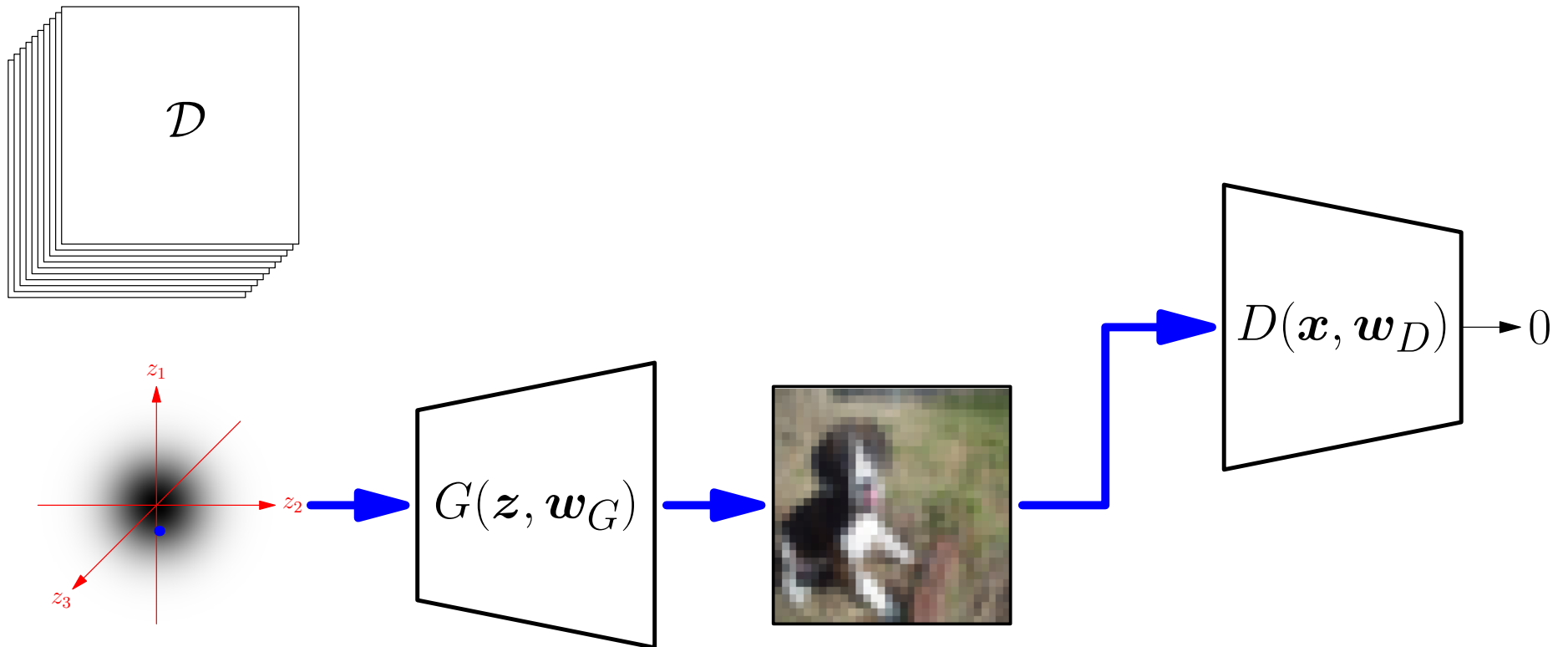
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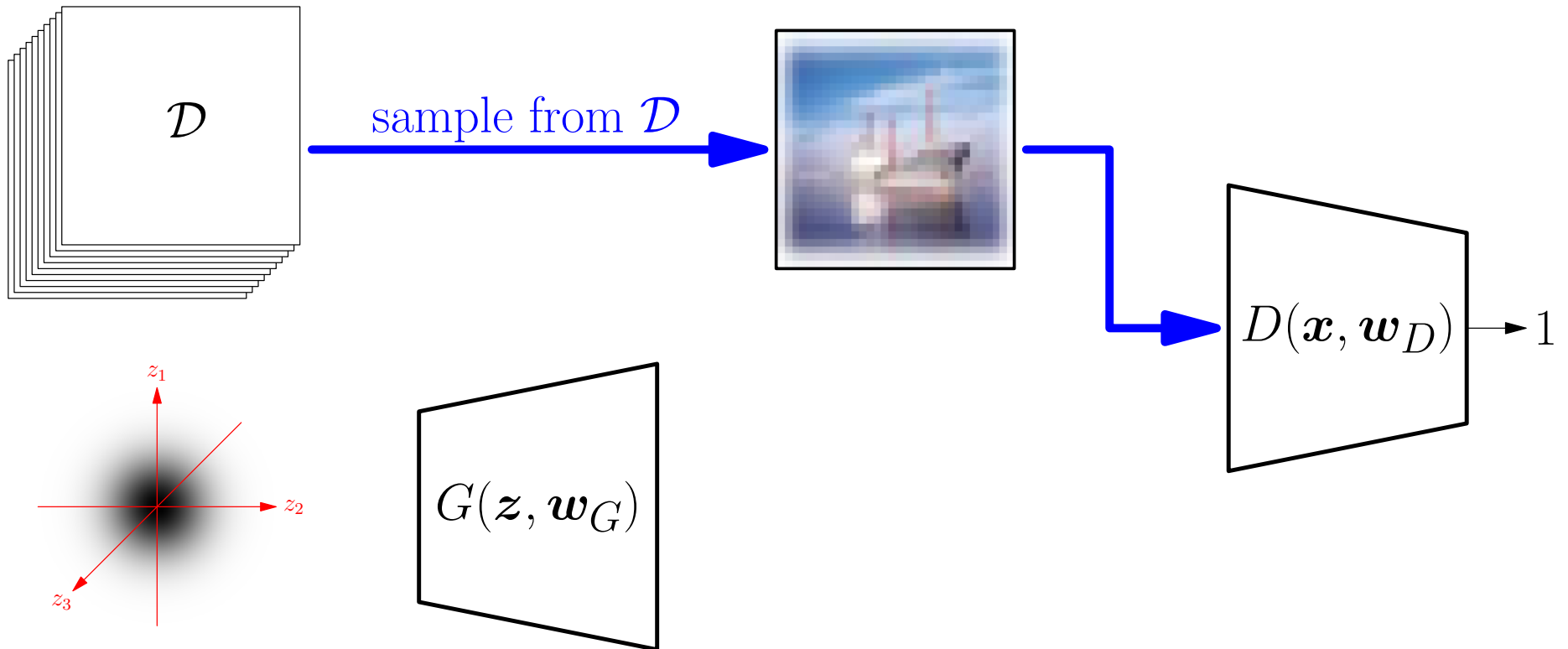
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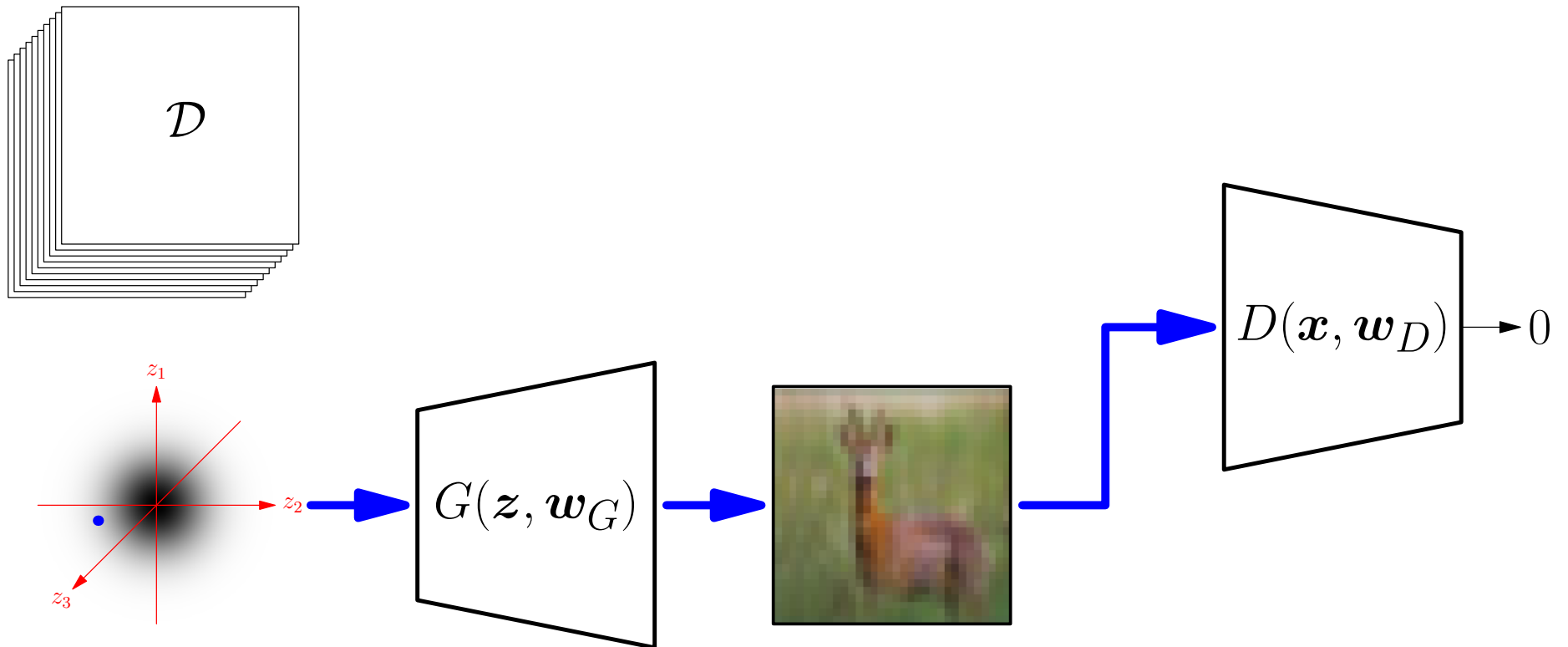
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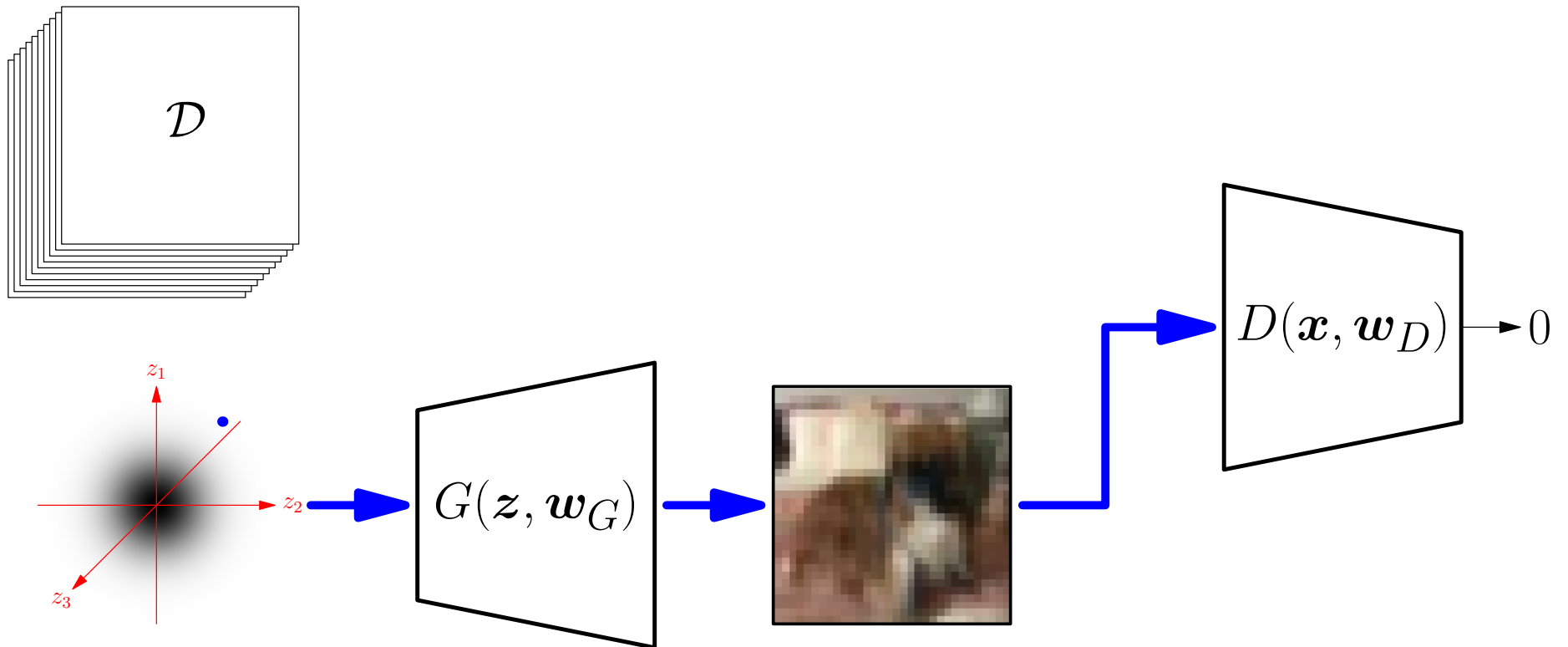
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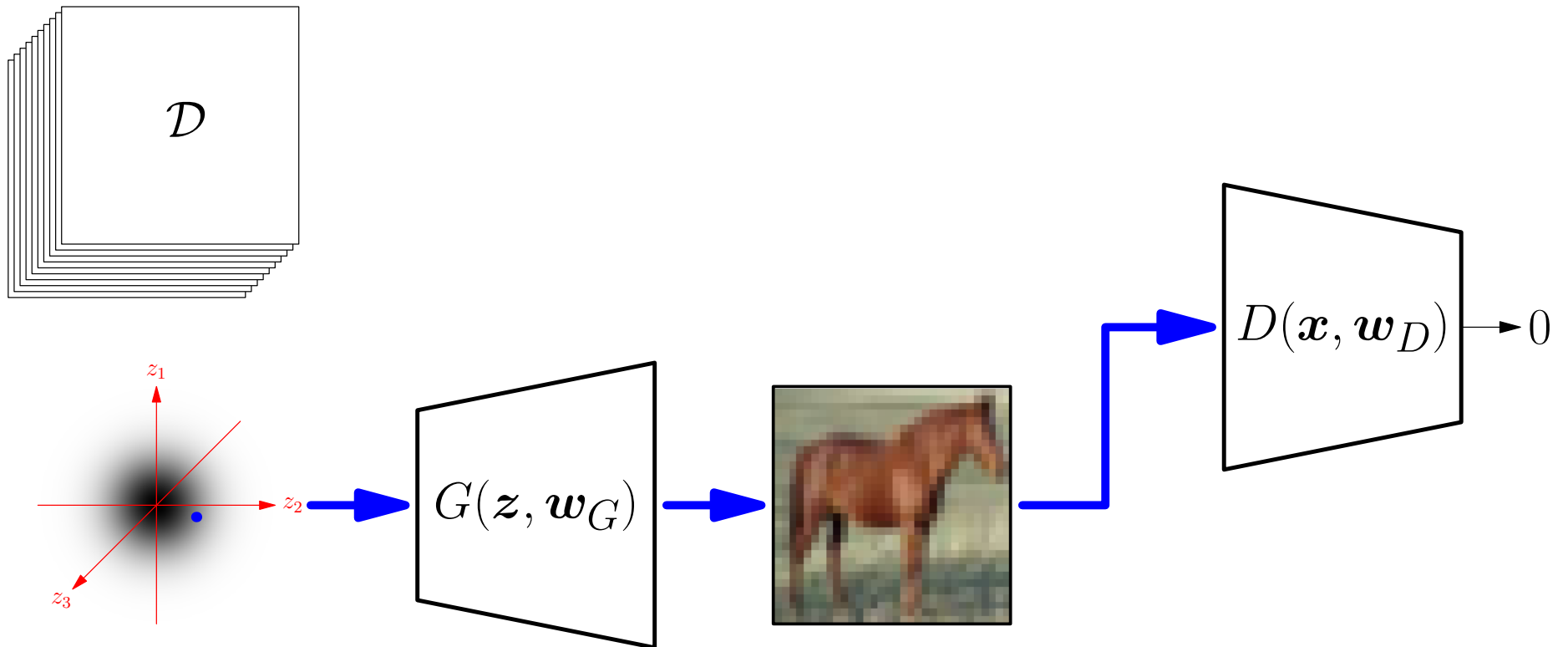
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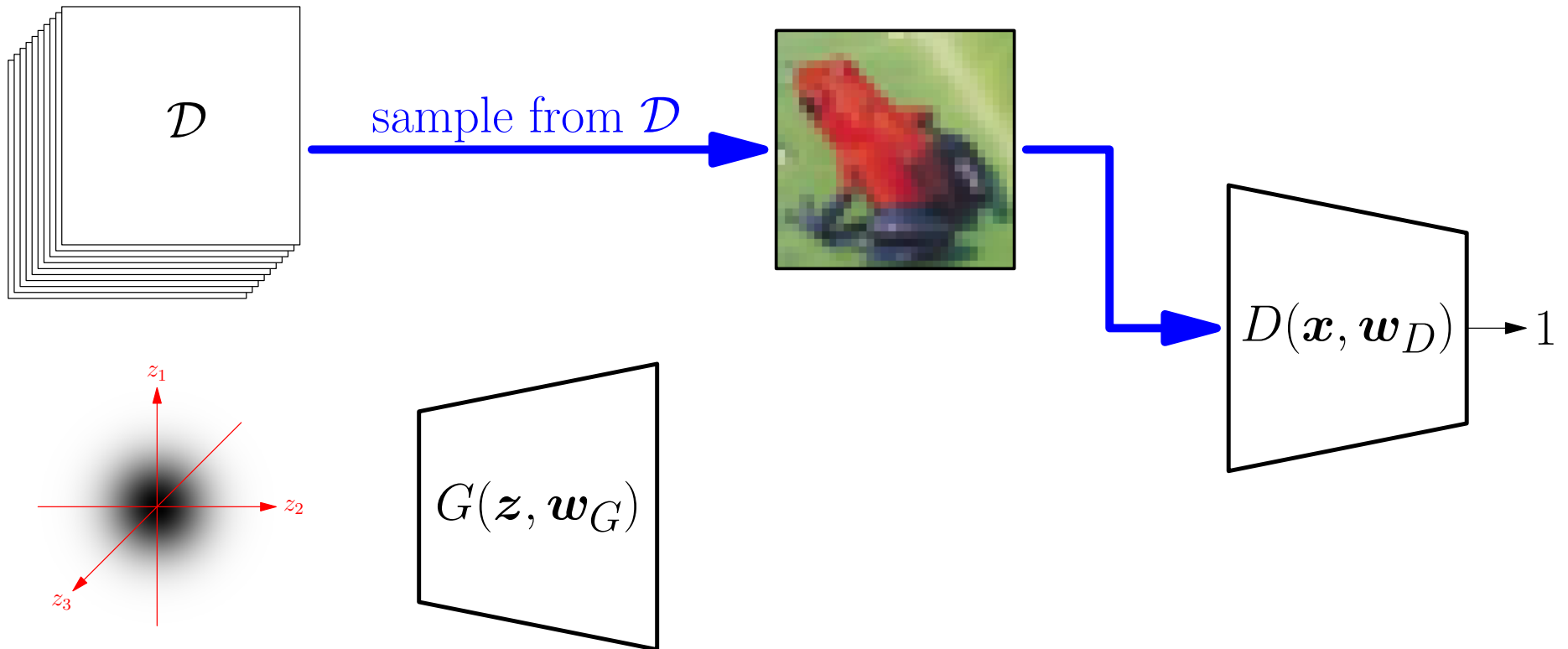
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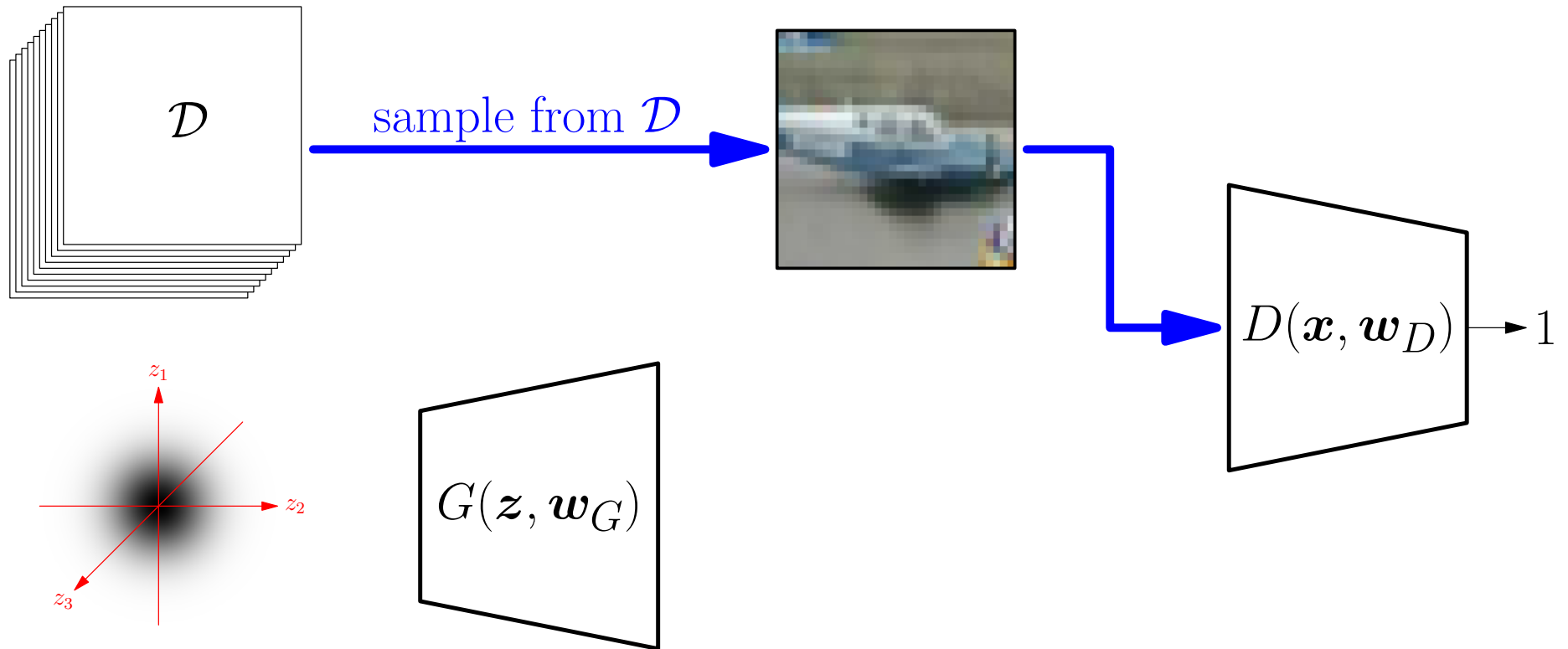
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How GANs Work



Training GANs

- The loss of the generator depends on its ability to trick the discriminator
- The loss of the discriminator depends on its ability not to be tricked
- We try to train the two networks simultaneously
- We hope that over time the generator produces better and better fakes

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Problems of GANs

- GANs are notoriously difficult to train
- The generator and discriminator training can decouple
- Often the discriminator becomes too good at correctly identifying the generated images
- Then there can be little gradient information to help the generator as every small change in parameters doesn't significantly change the discriminator decision
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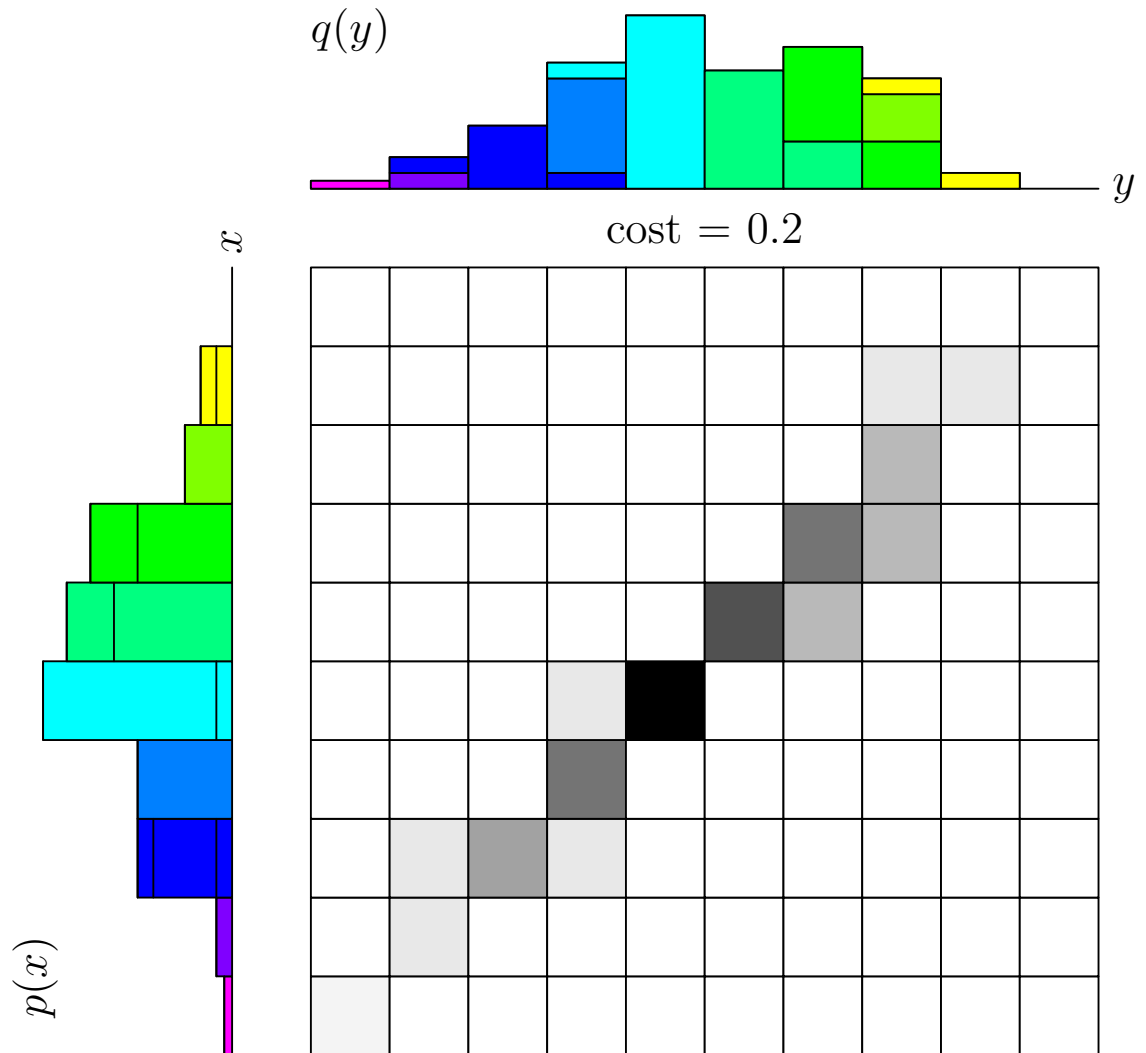
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Outline

1. GANs
2. **Wasserstein Distance**
3. Wasserstein GANs



Measuring Distances Between Distributions

- In many machine learning tasks we want to minimise the distance between two probability distributions
- This requires that we can measure distances between probability distributions
- One prominent measure is the Kullback-Leibler or KL divergence

$$\text{KL}(p||q) = \int p(\mathbf{x}) \log\left(\frac{p(\mathbf{x})}{q(\mathbf{y})}\right) d\mathbf{x}$$

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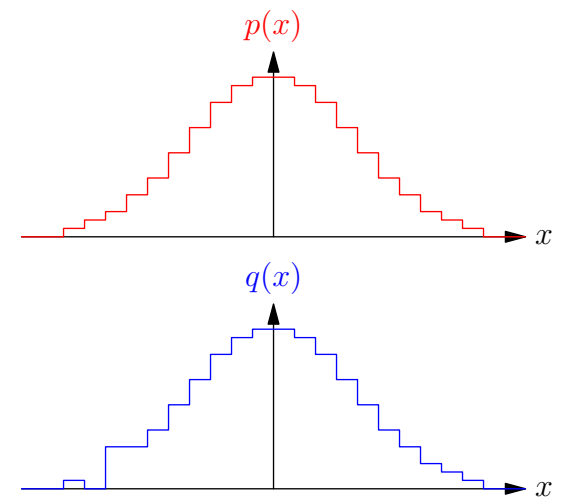
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Trouble with KL

- KL-divergences are non-negative quantities that are minimised when the two probability distributions are the same
- They are not distances (they aren't symmetric and they don't satisfy the triangular inequality)

We don't really care about this, but what

- we do care about is that if $q(\mathbf{x}) = 0$ when $p(\mathbf{x}) \neq 0$ then $\log\left(\frac{p(\mathbf{x})}{q(\mathbf{y})}\right)$ diverges



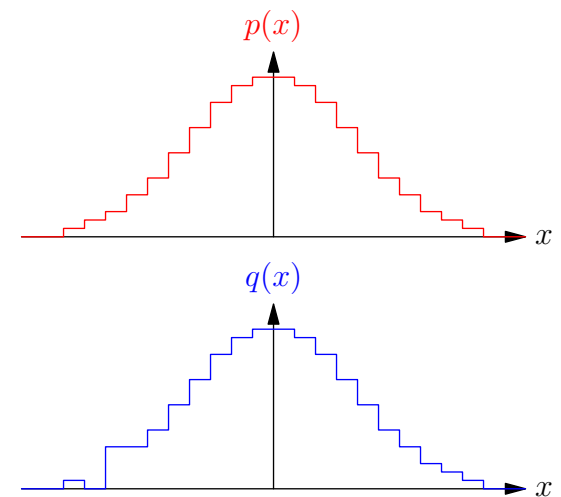
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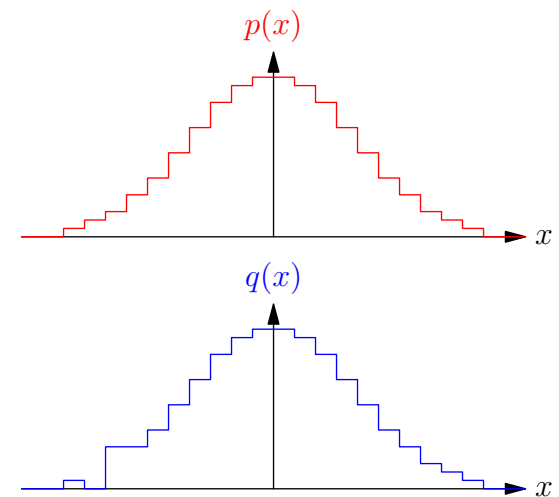


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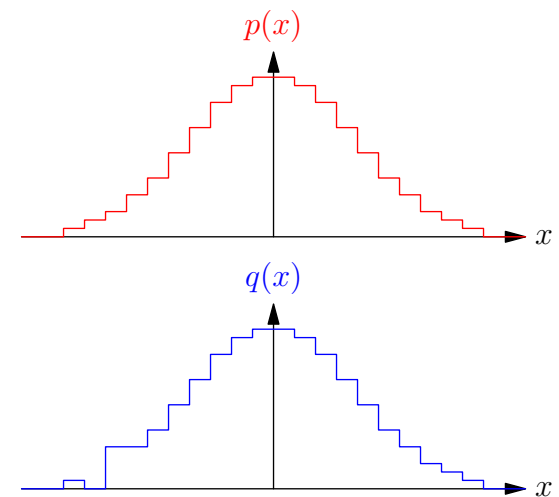
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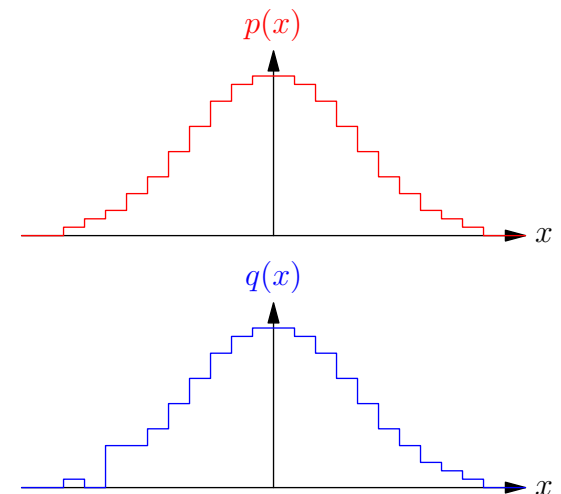


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Wasserstein Distance

- A more benign measure of the differences between two probability functions is the **Wasserstein** or **Earth Moving** distance

- This is a true distance, but more importantly, it measures distance in a very natural way so that distributions that are close has a small Wasserstein distance

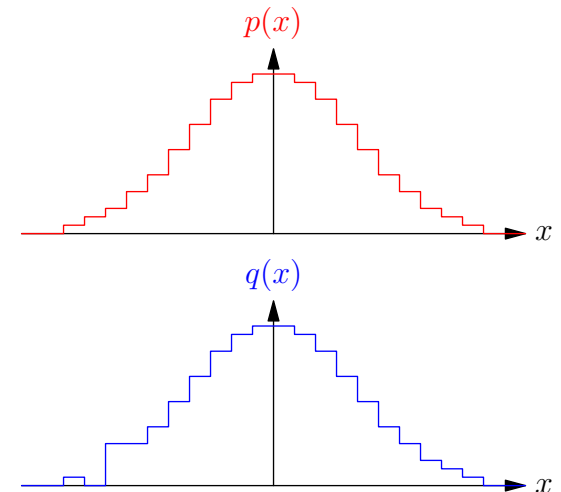


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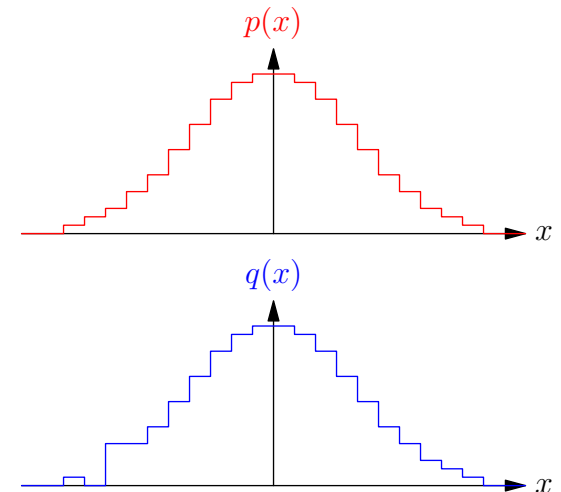


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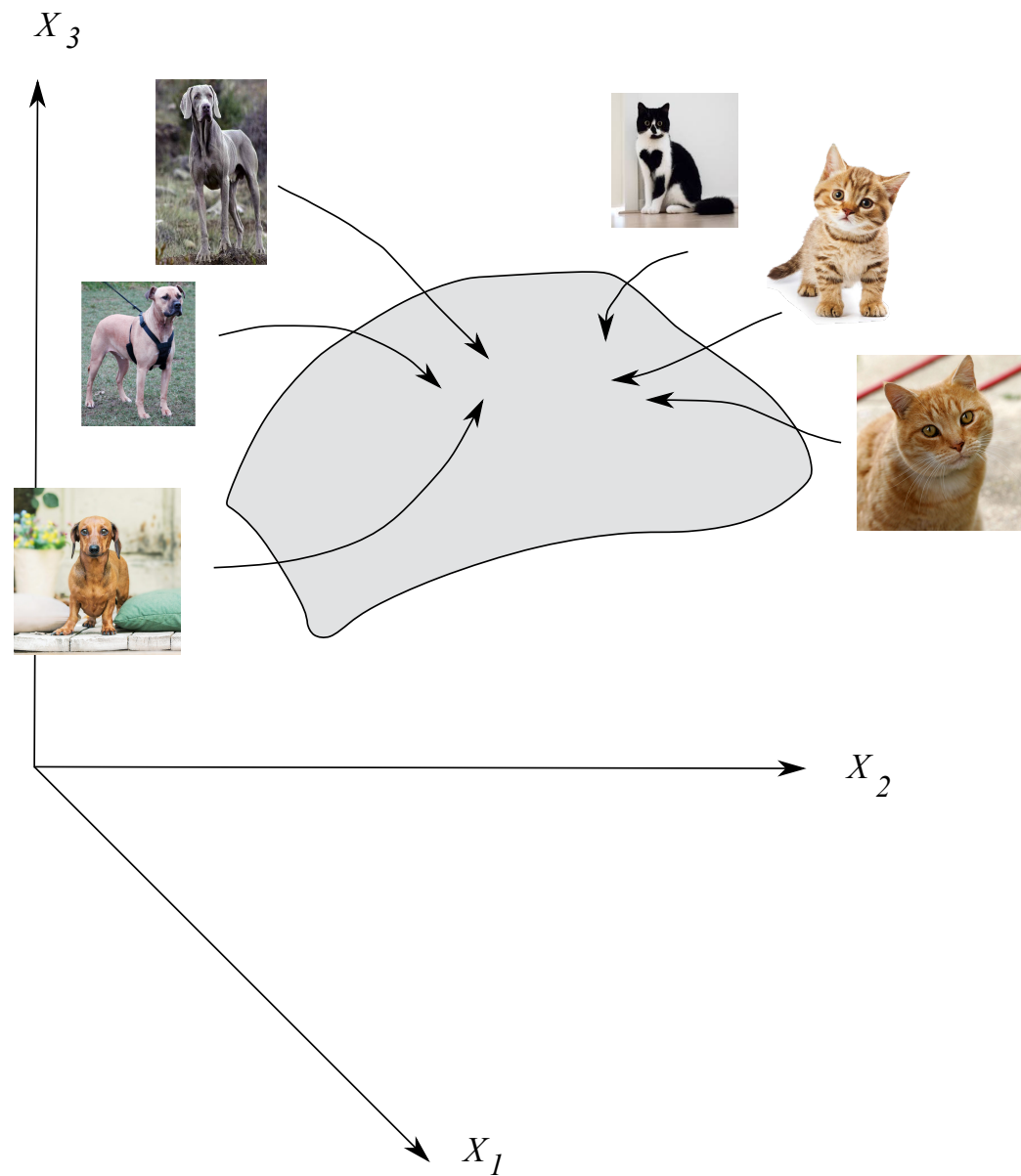
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High Probability Manifold



Transportation Policy

- But how do we formalise the Wasserstein distance?
- A good place to start is to define a transportation policy $\gamma(\mathbf{x}, \mathbf{y})$ with

$$\int \gamma(\mathbf{x}, \mathbf{y}) d\mathbf{y} = p(\mathbf{x}) \qquad \int \gamma(\mathbf{x}, \mathbf{y}) d\mathbf{x} = q(\mathbf{y})$$

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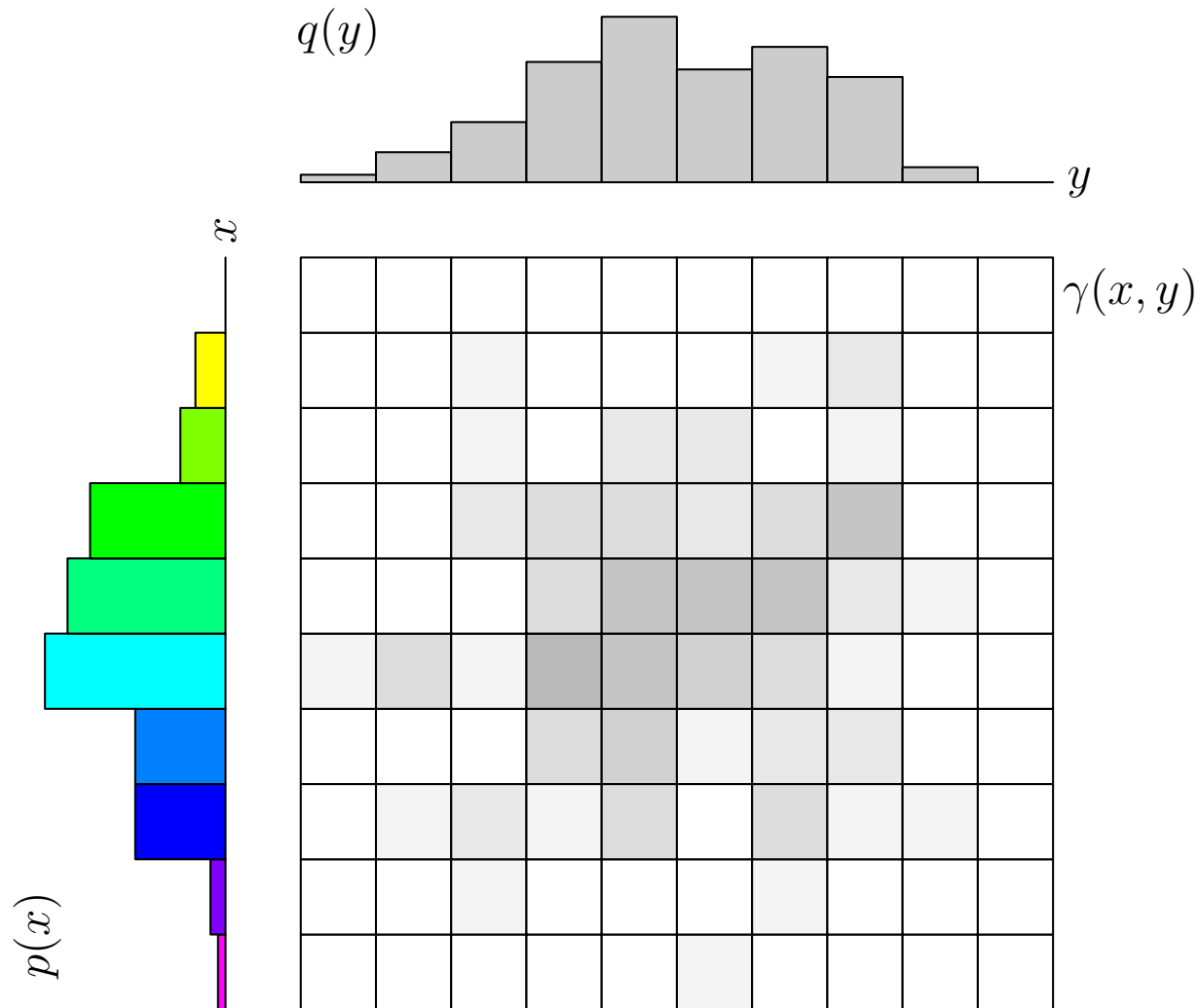
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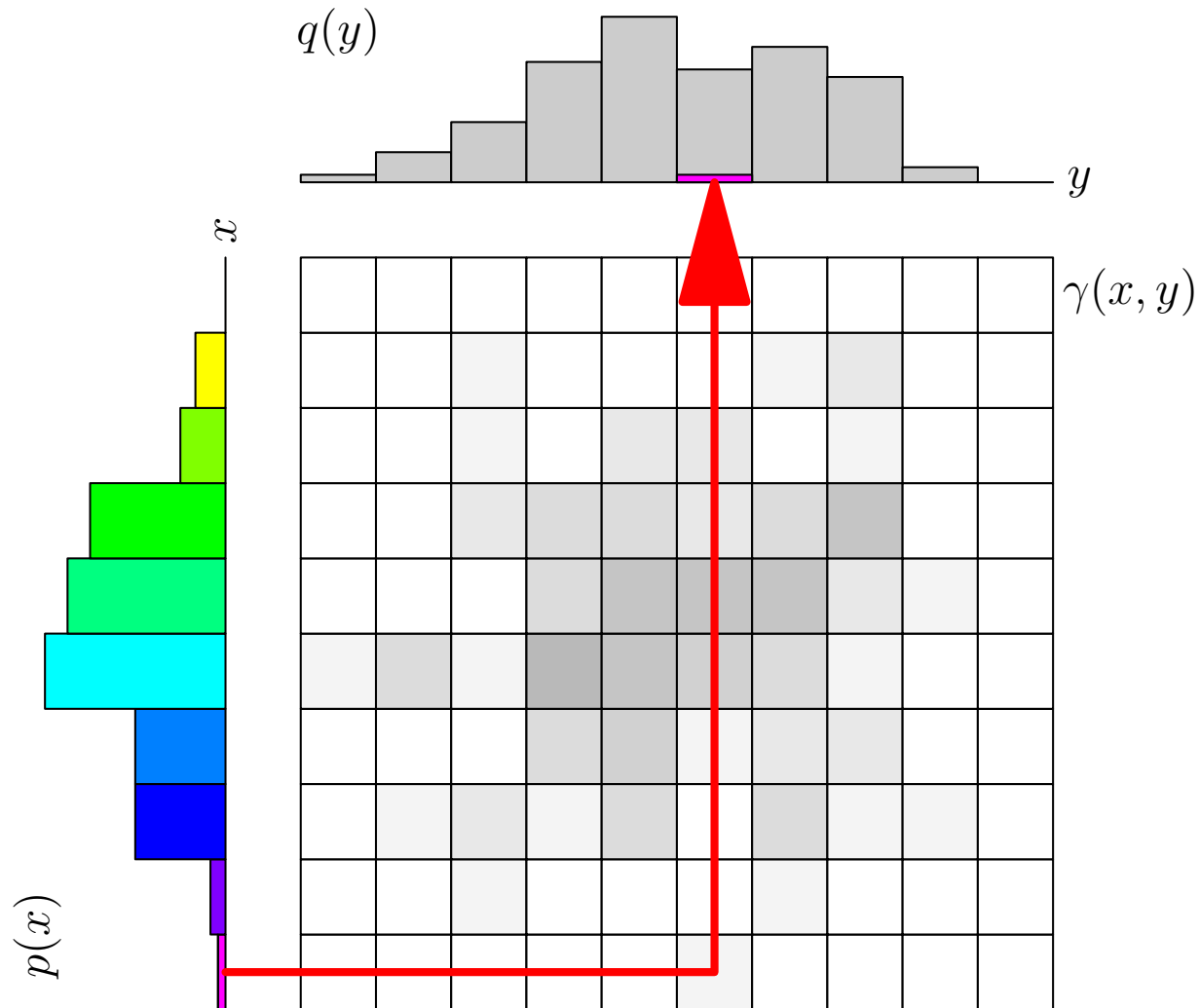
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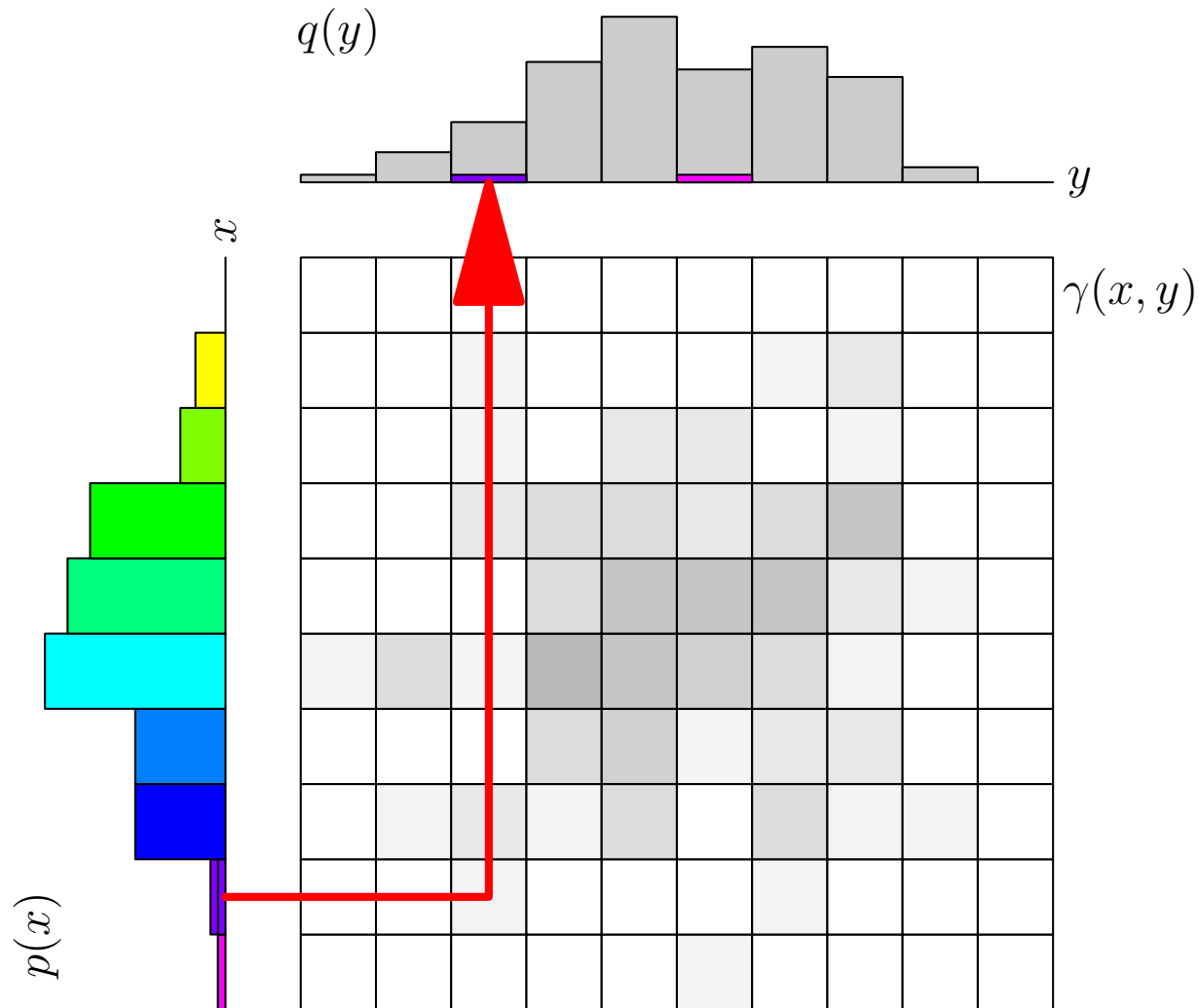
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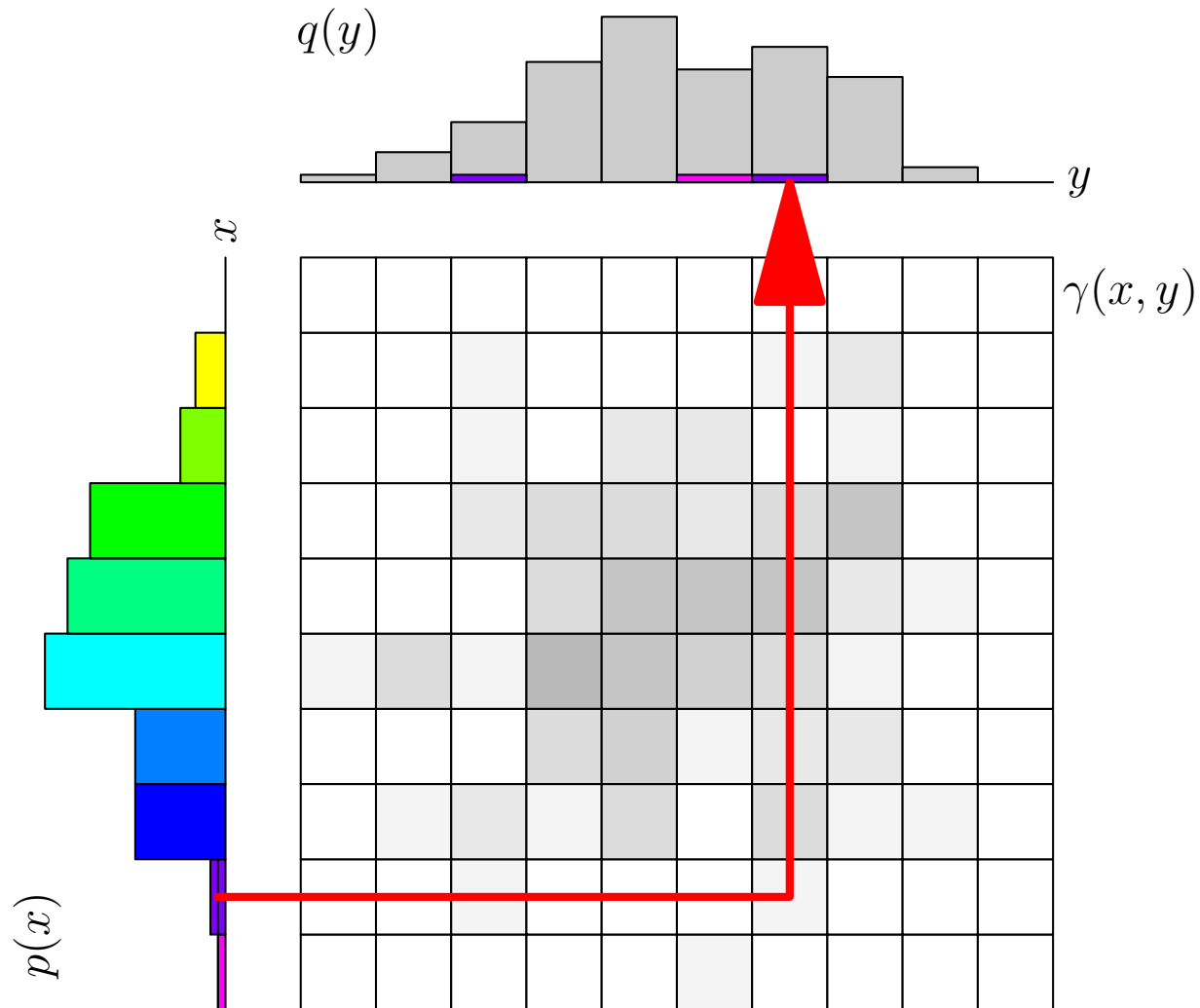
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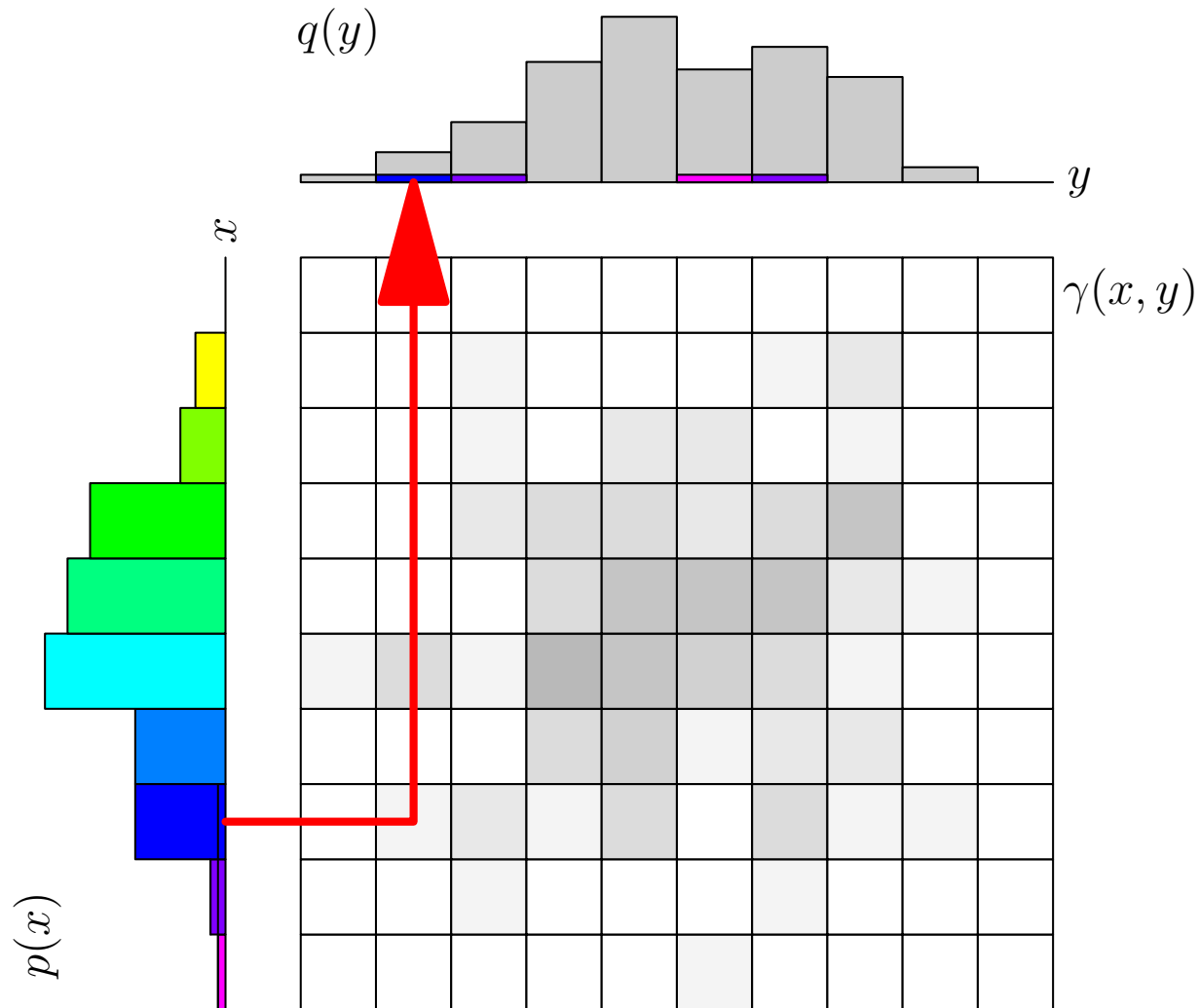
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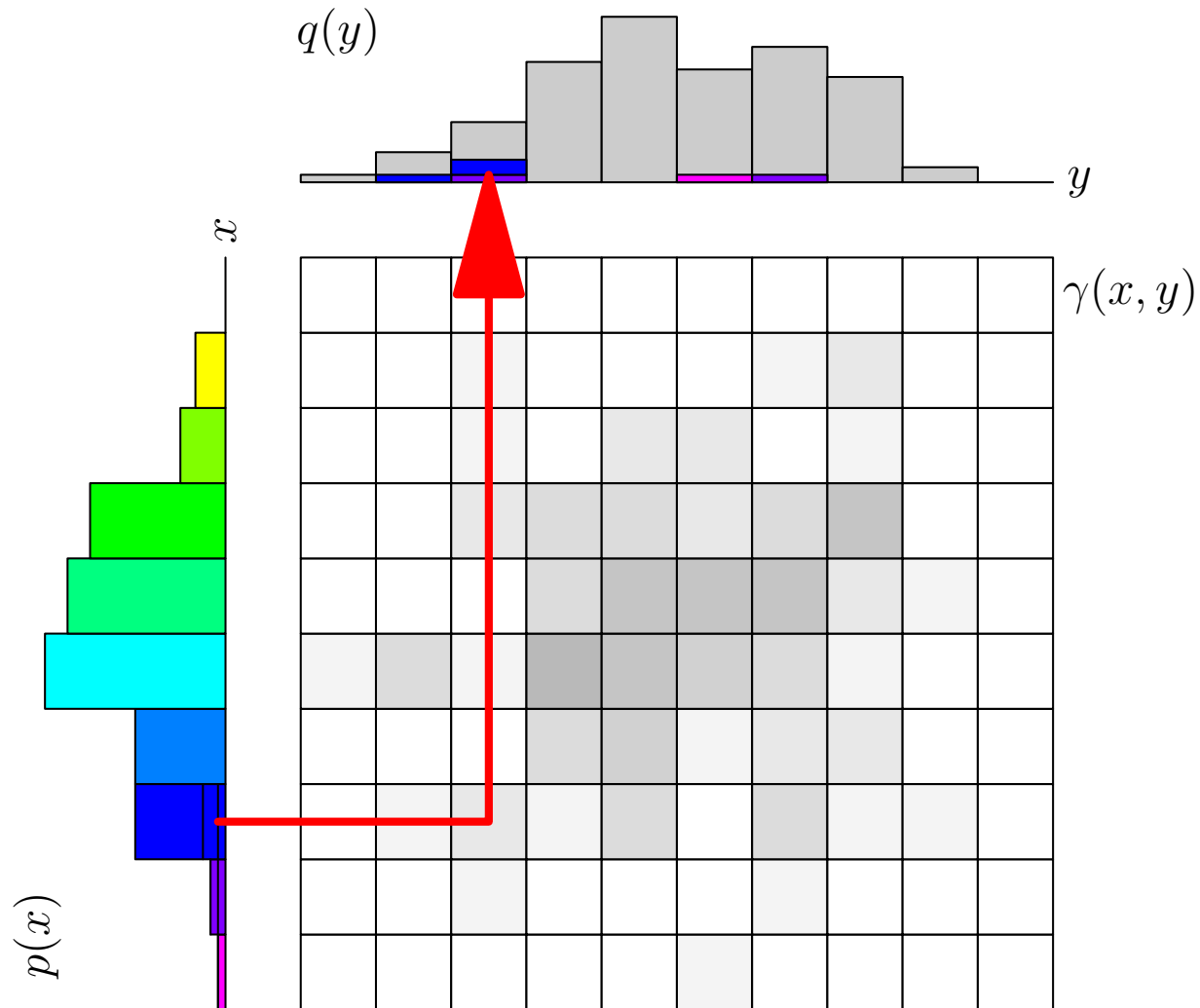
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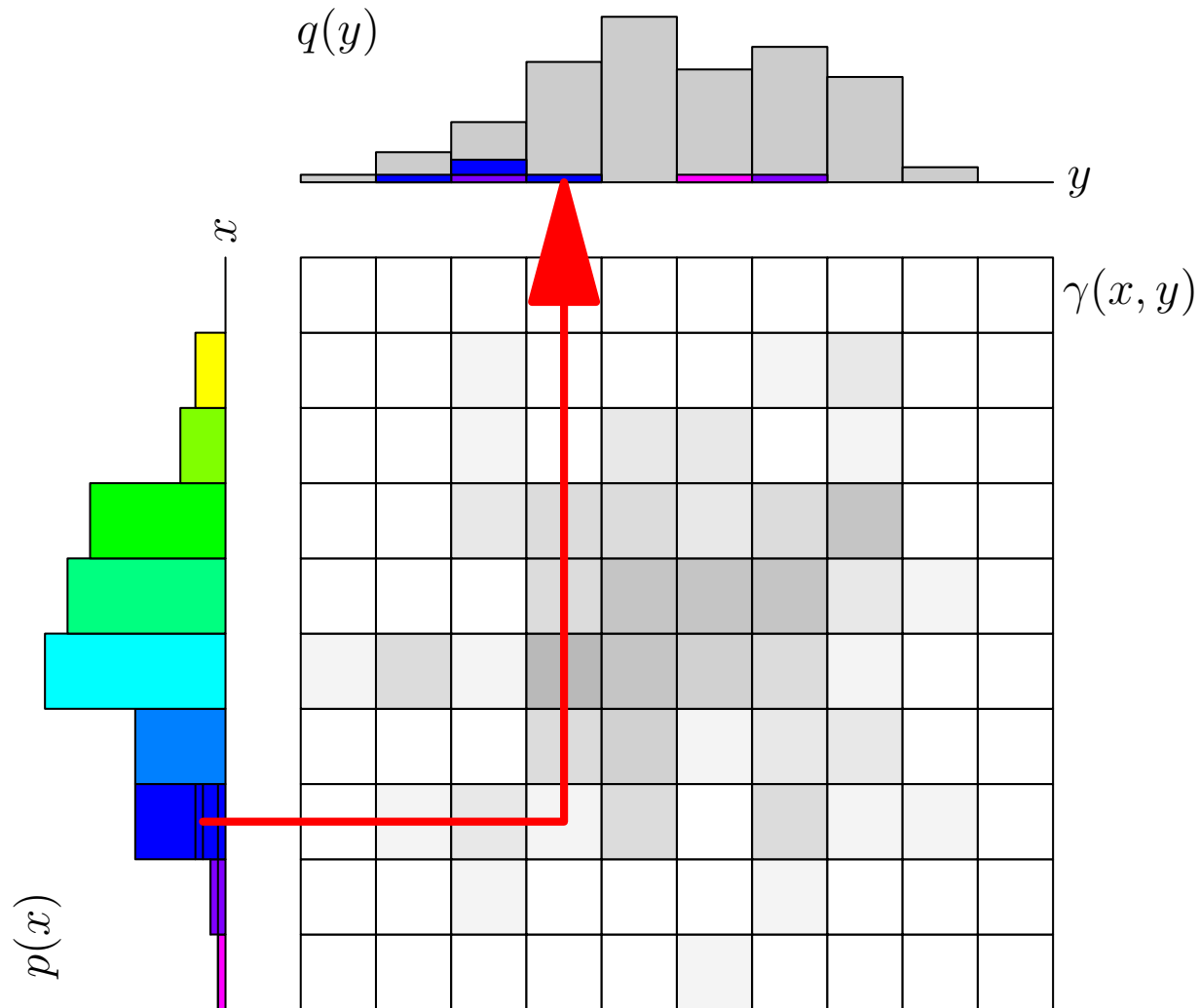
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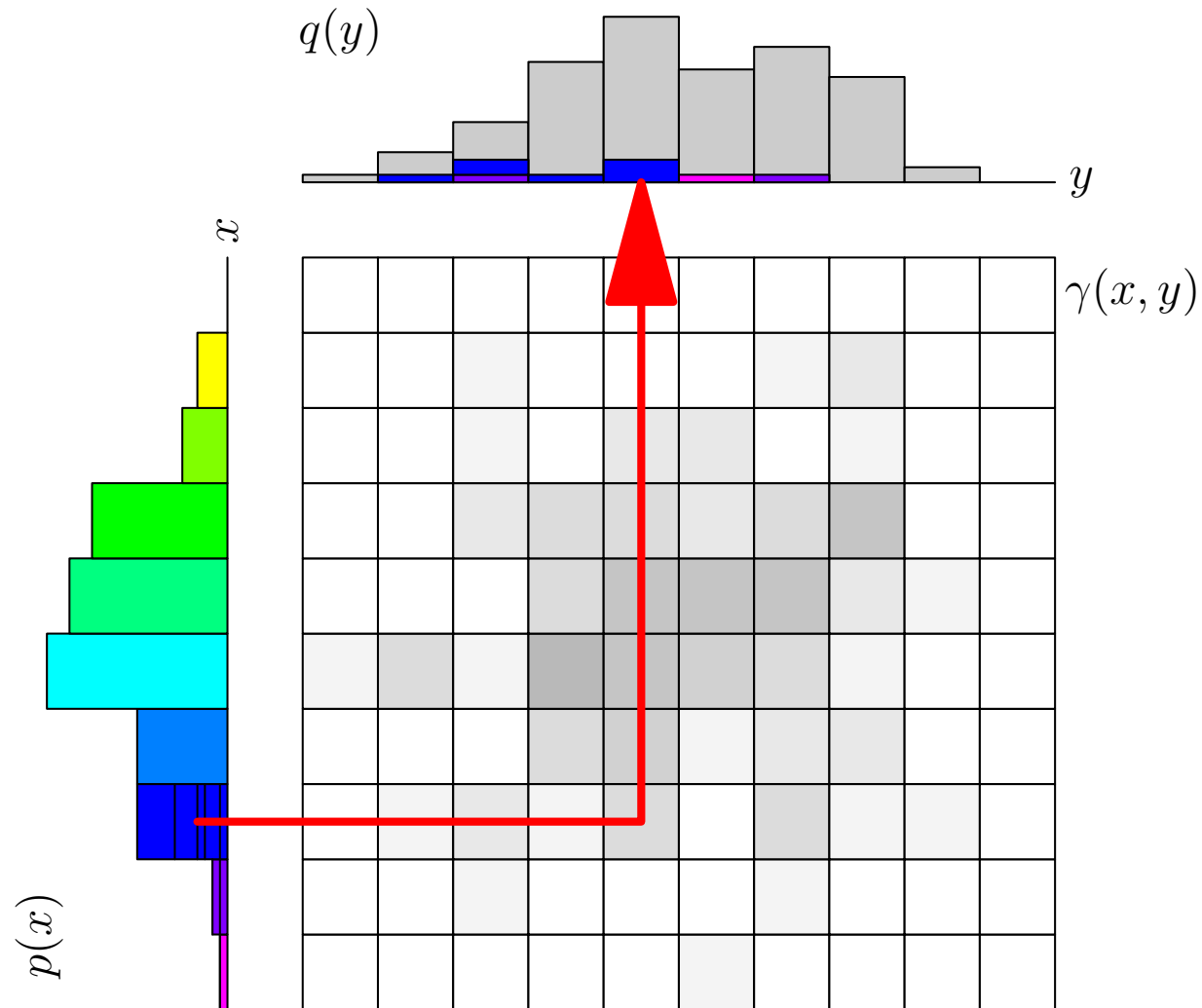
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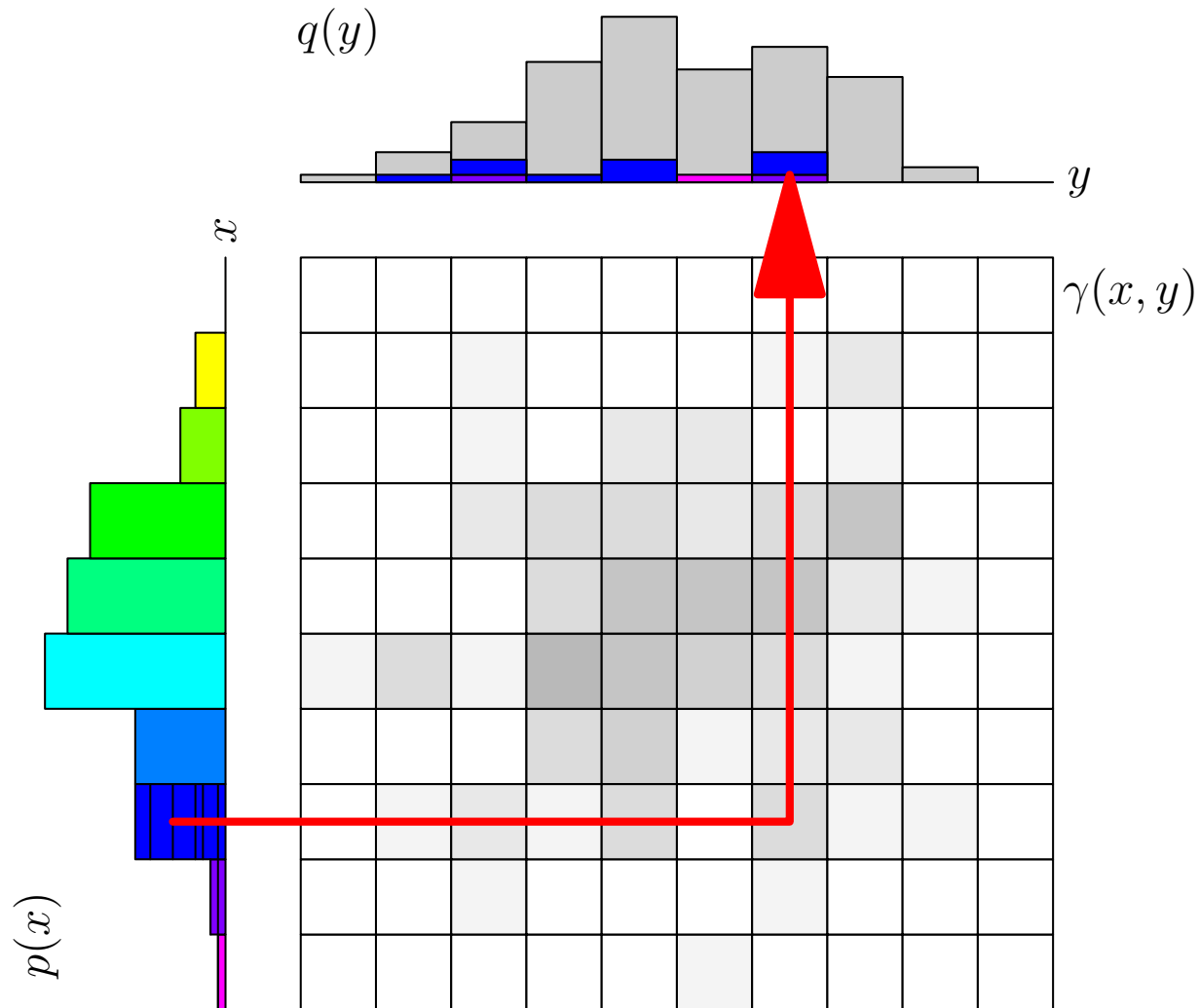
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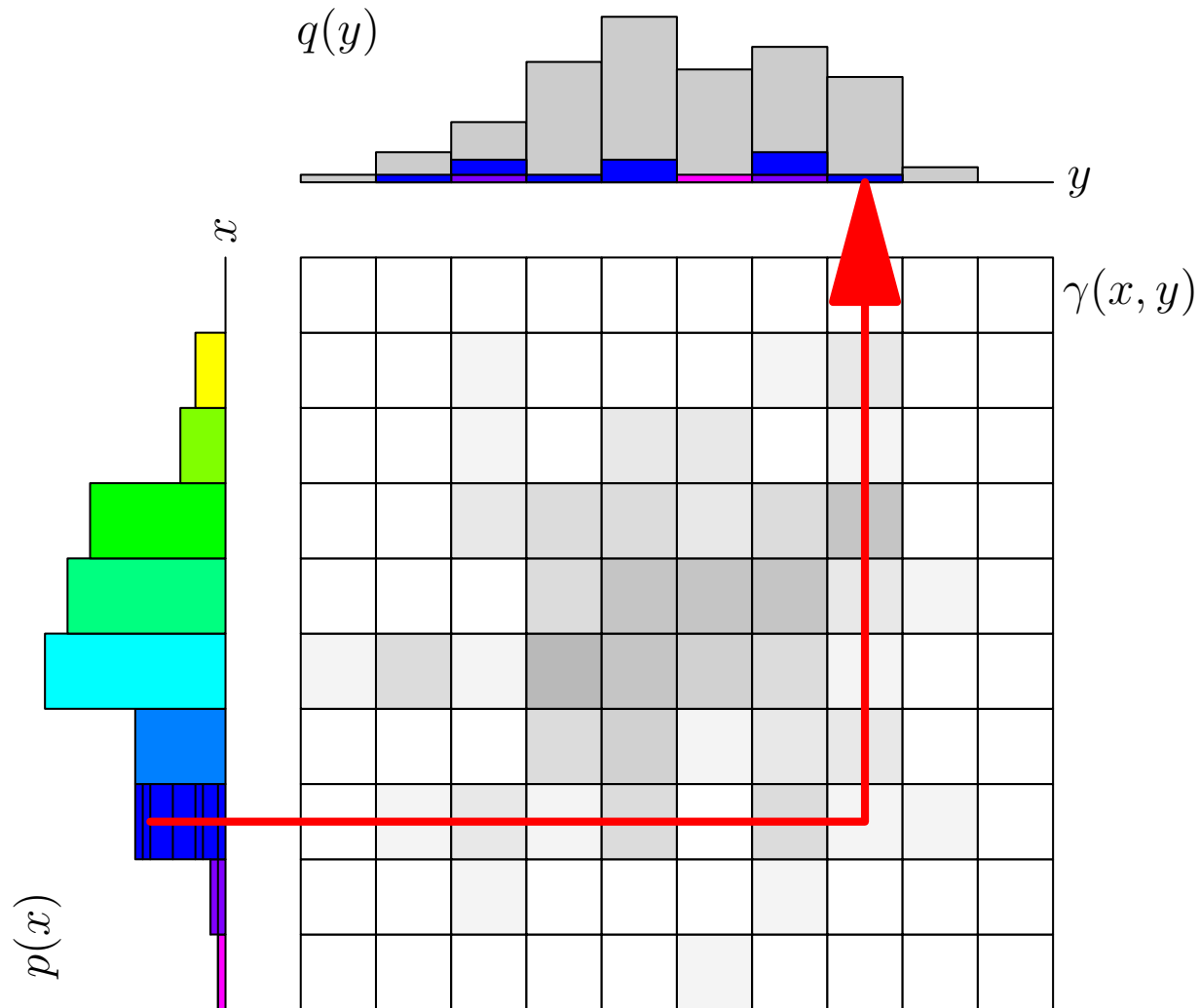
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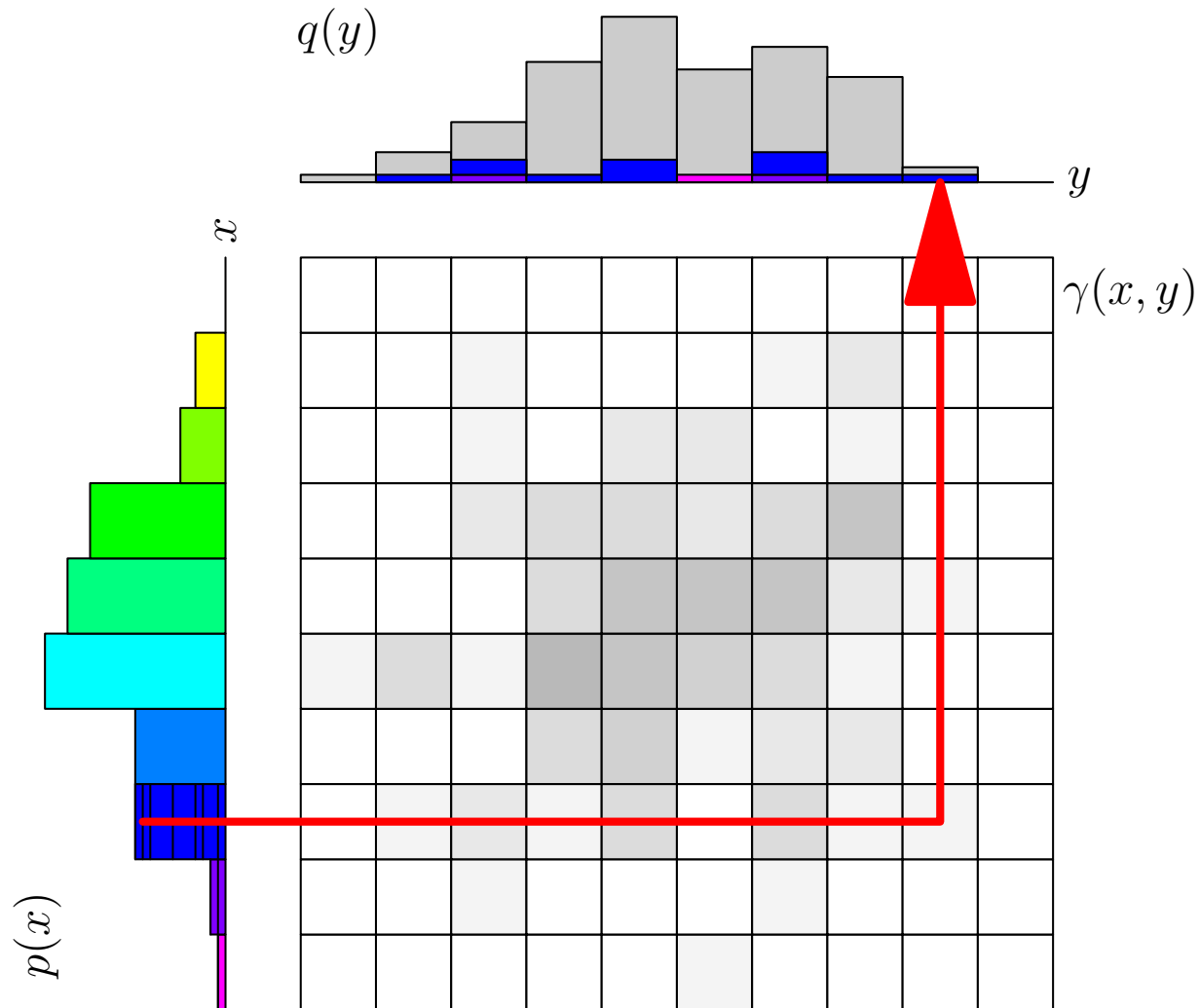
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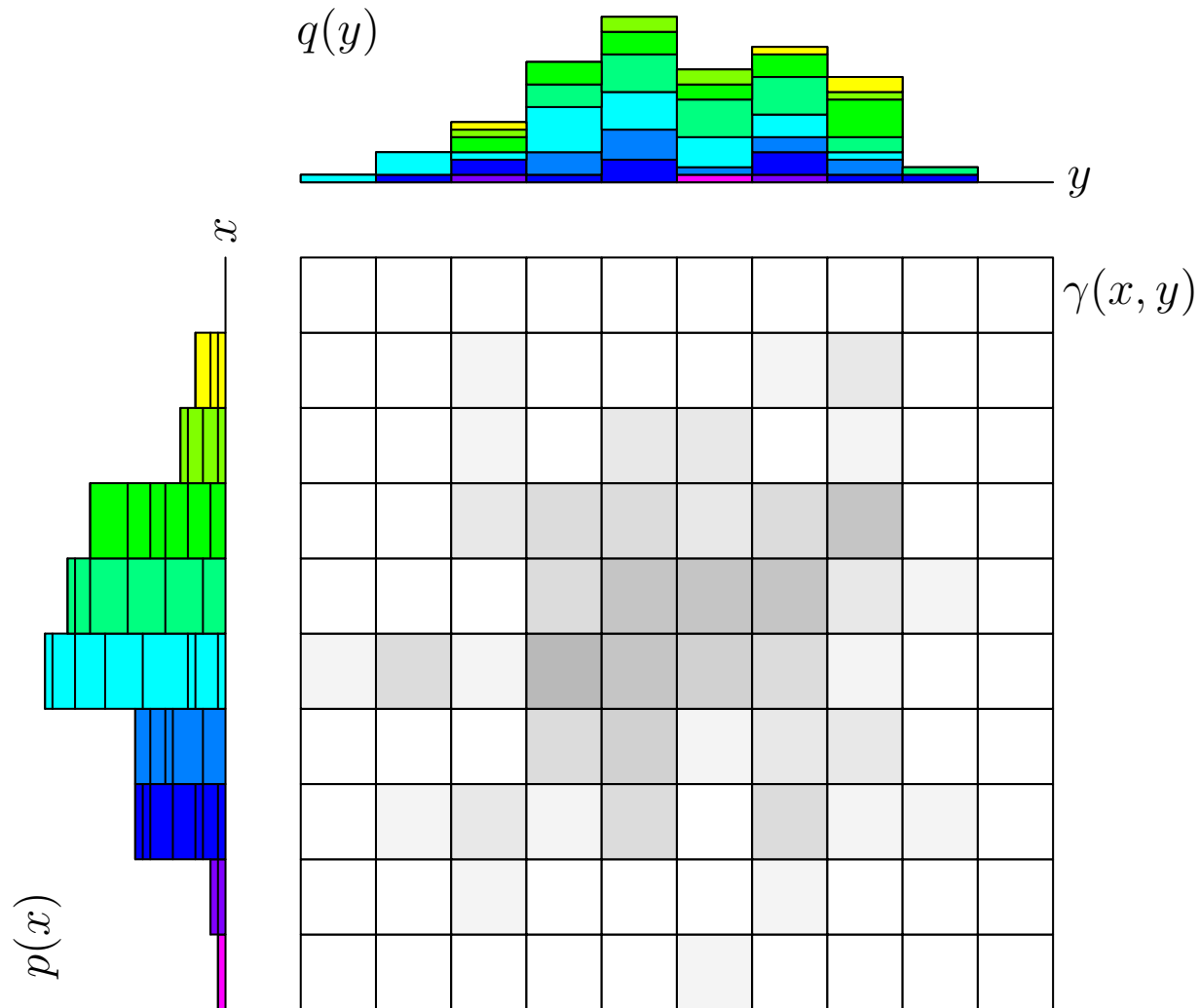
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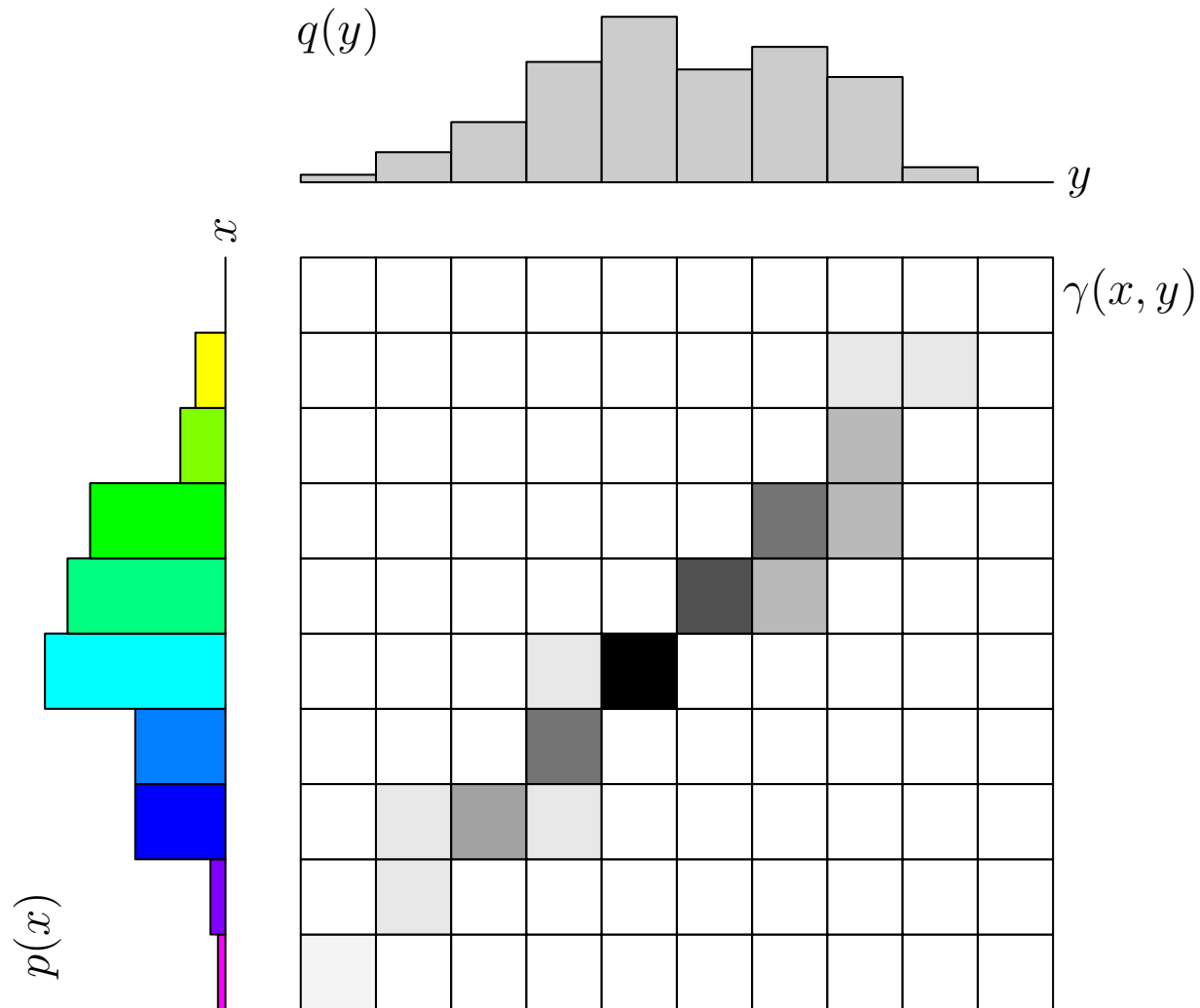
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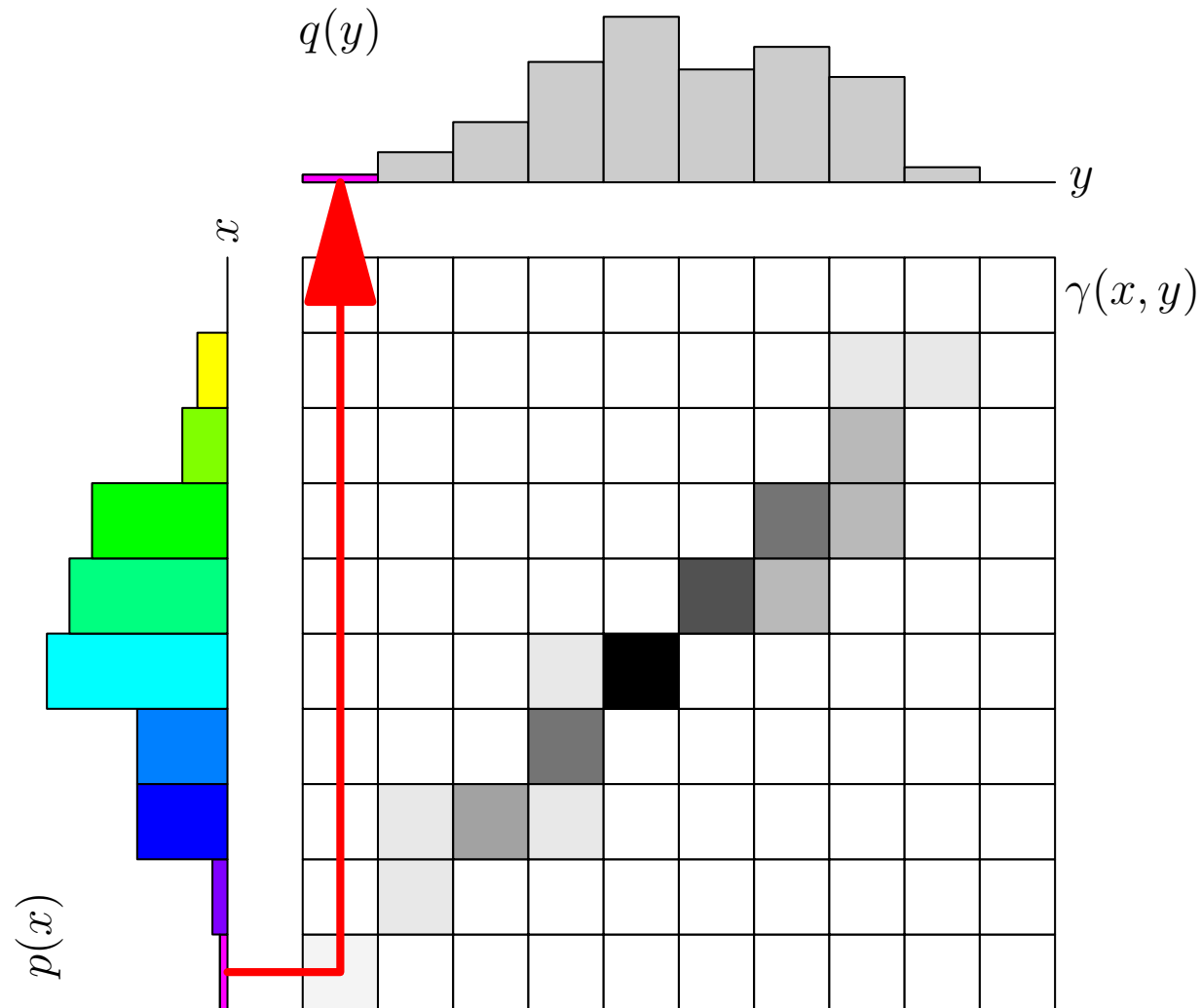
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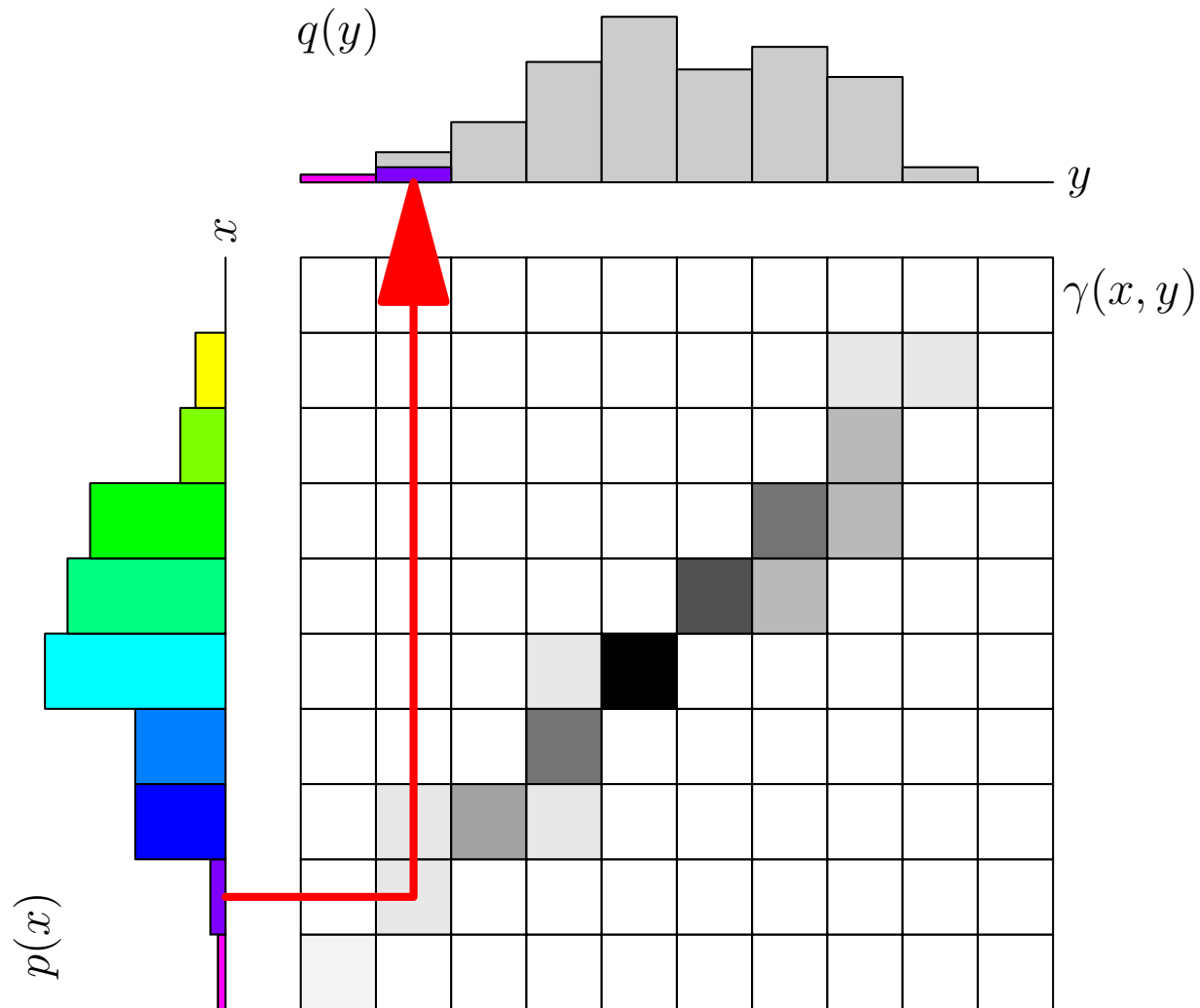
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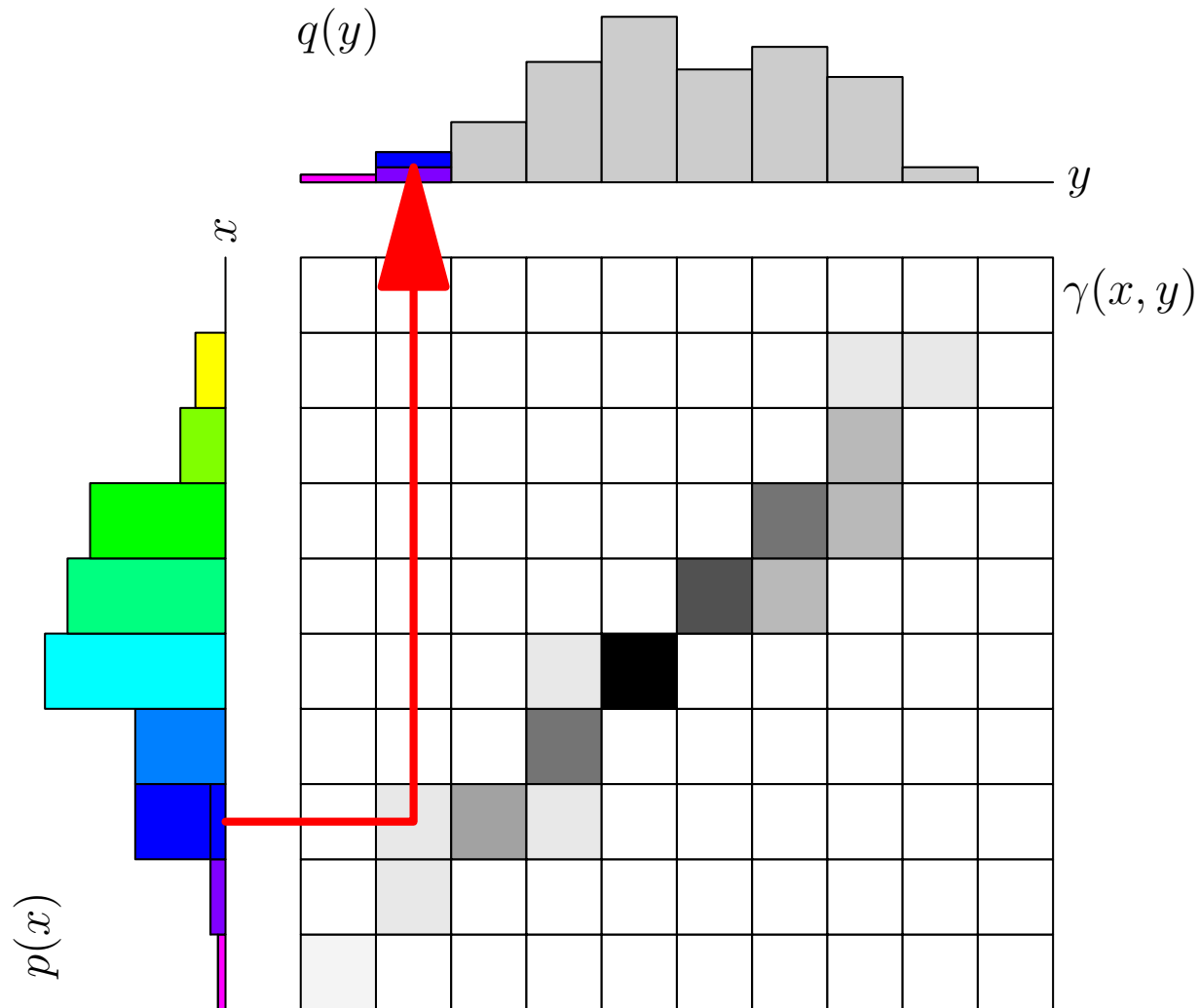
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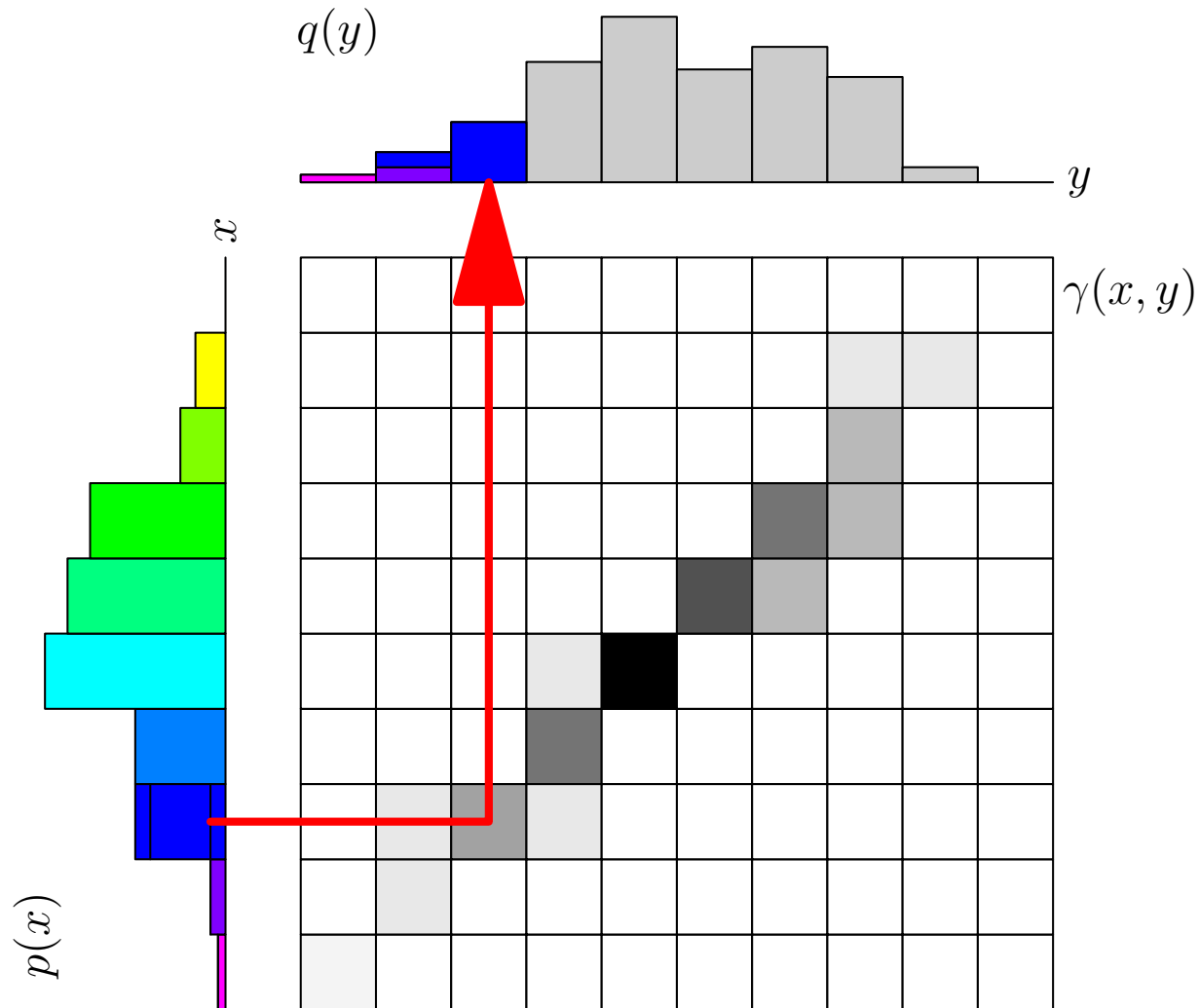
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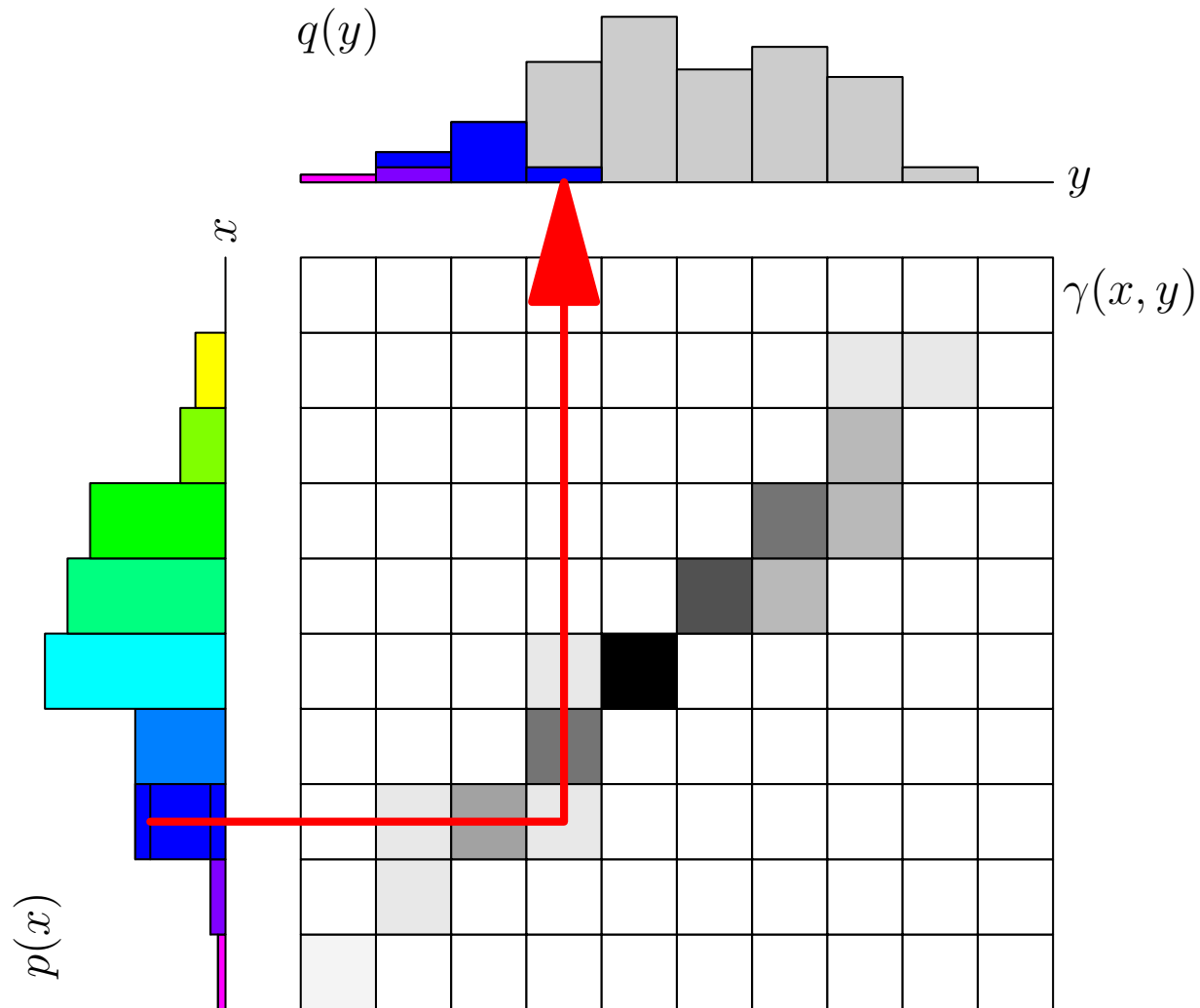
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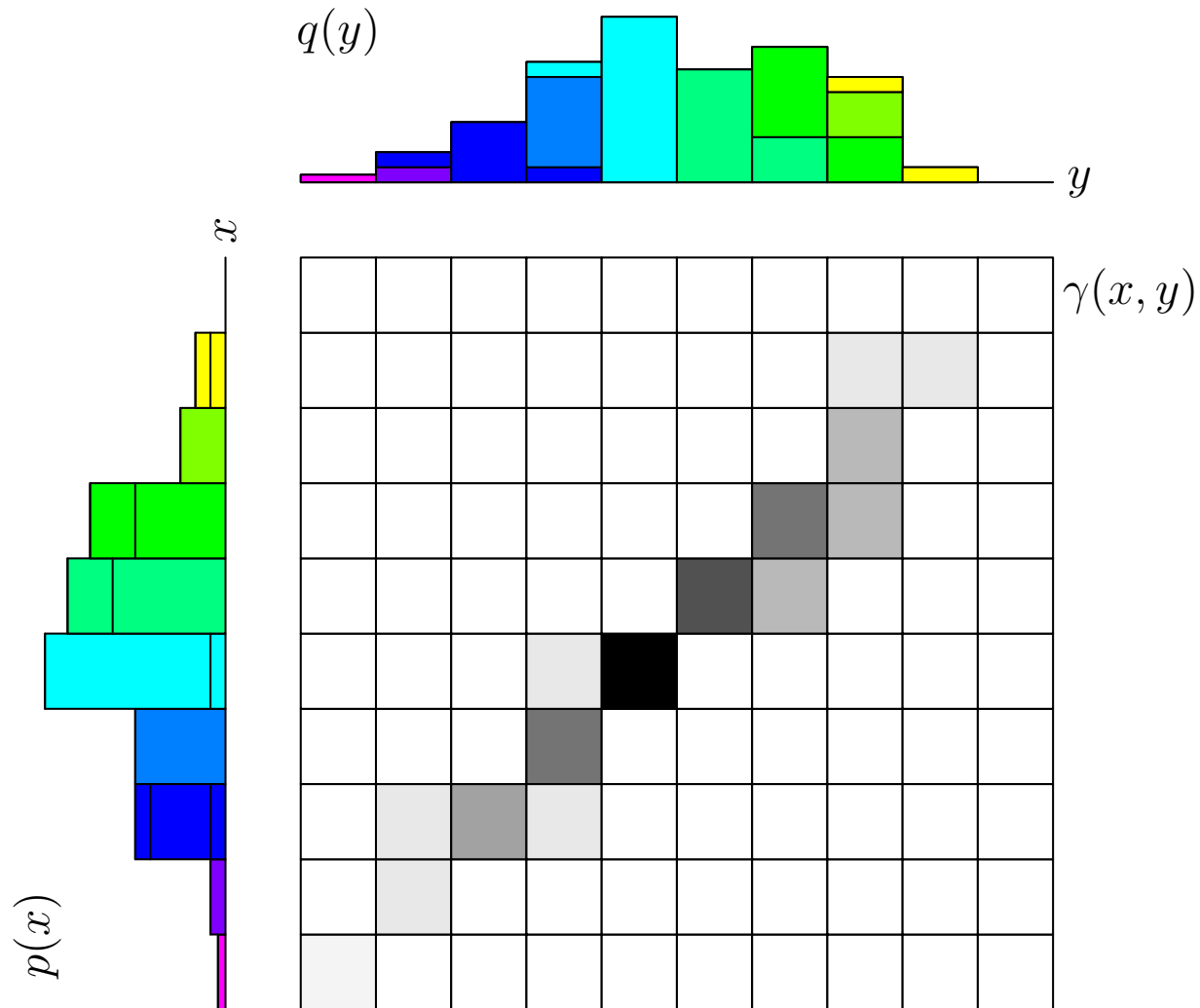
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Transportation Policy



The Cost of Transport

- We want to choose the transportation policy that minimises the amount of probability mass we need to move
- Let $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ be a distance measure then the cost of a transportation policy is

$$C(\gamma) = \int \int d(\mathbf{x}, \mathbf{y}) \gamma(\mathbf{x}, \mathbf{y}) d\mathbf{x} d\mathbf{y}$$

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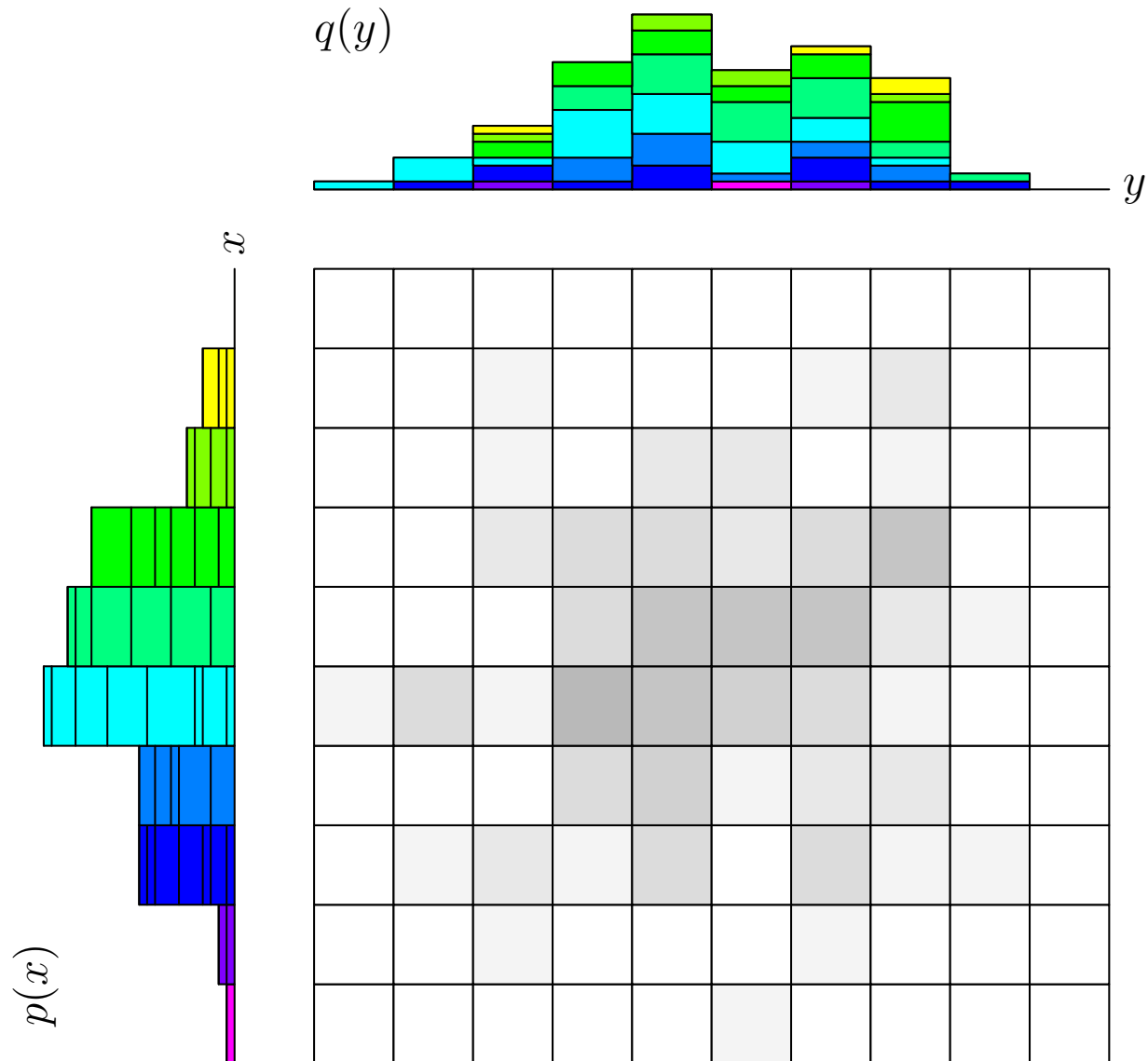
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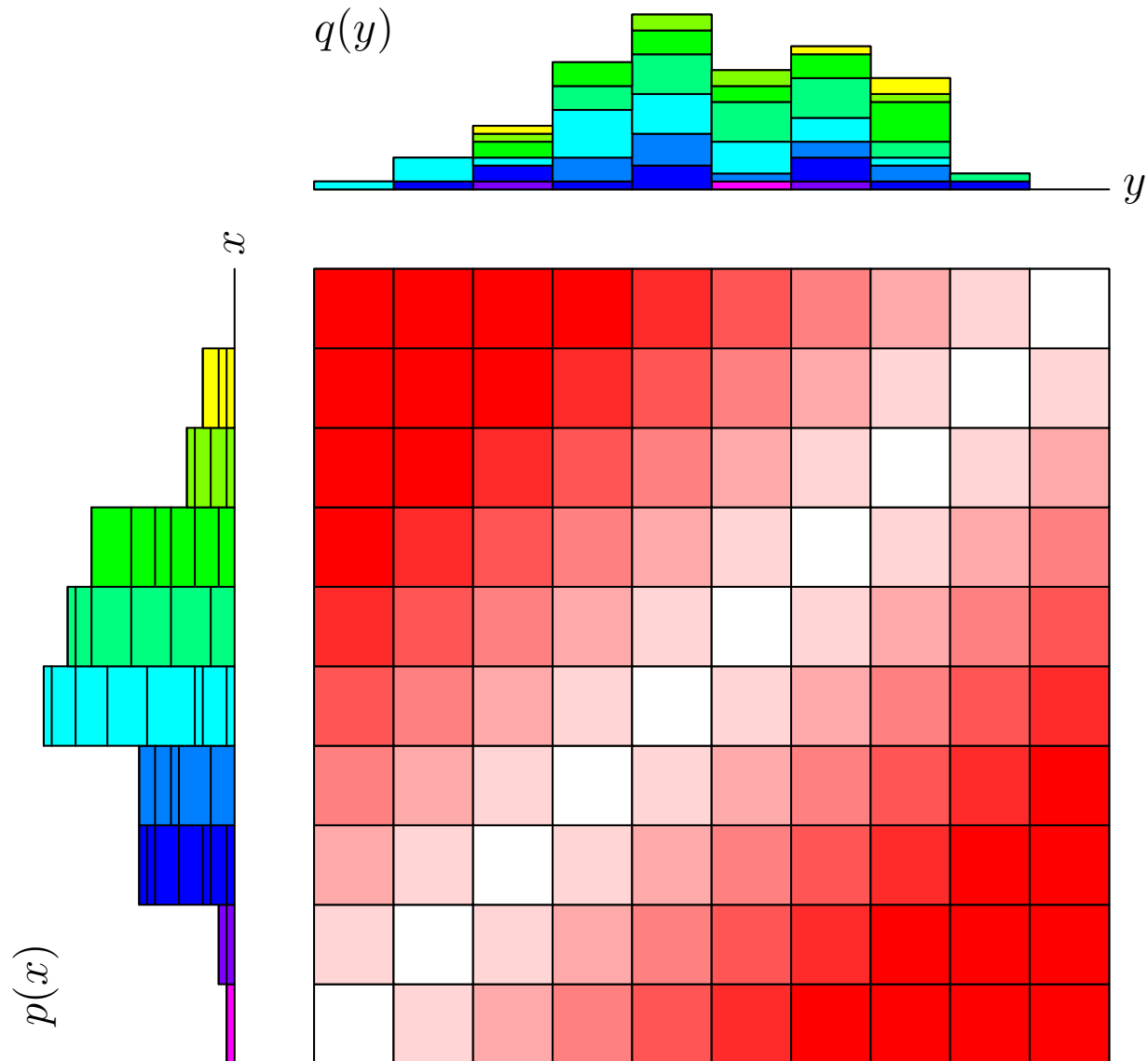
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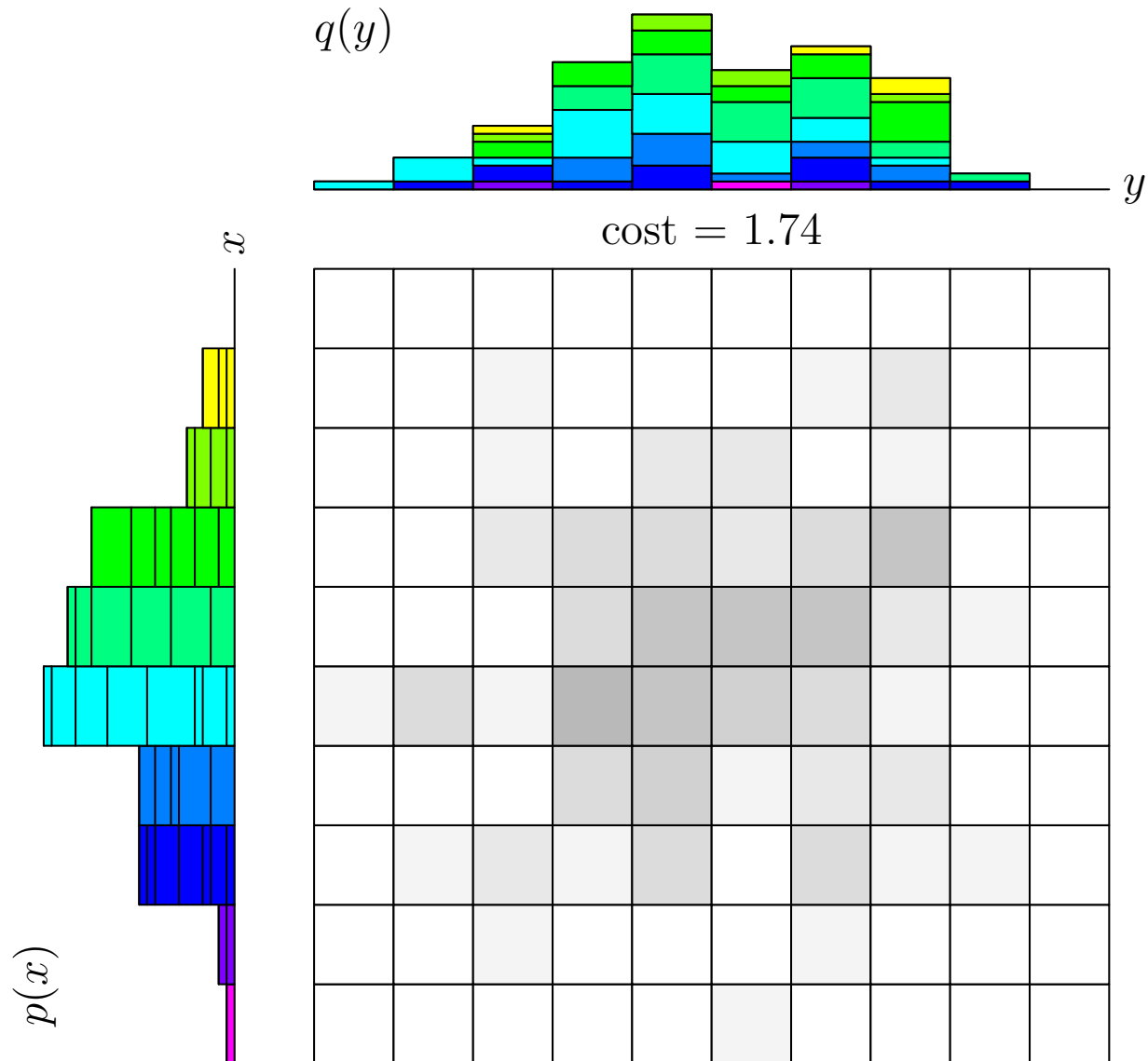
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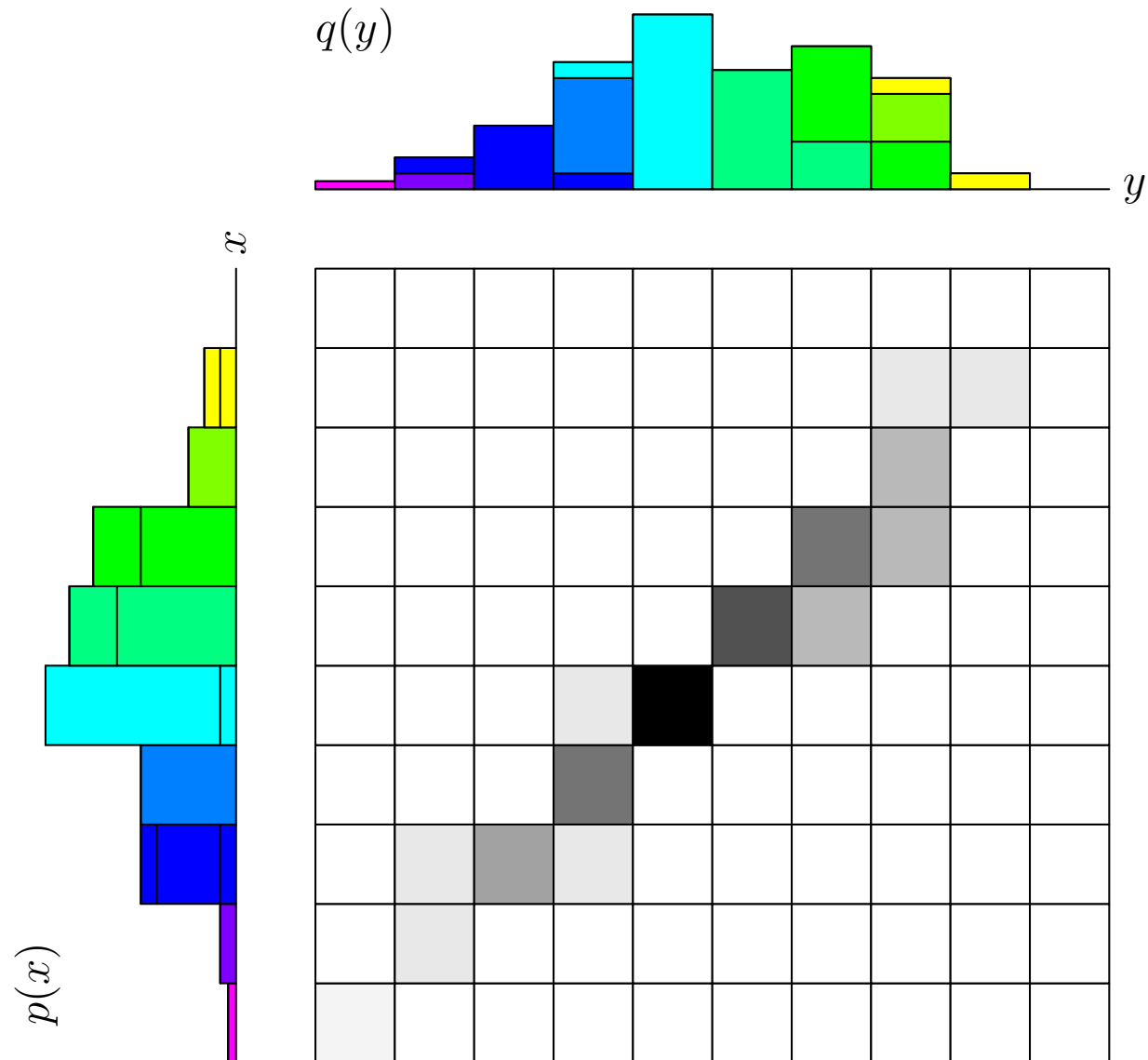
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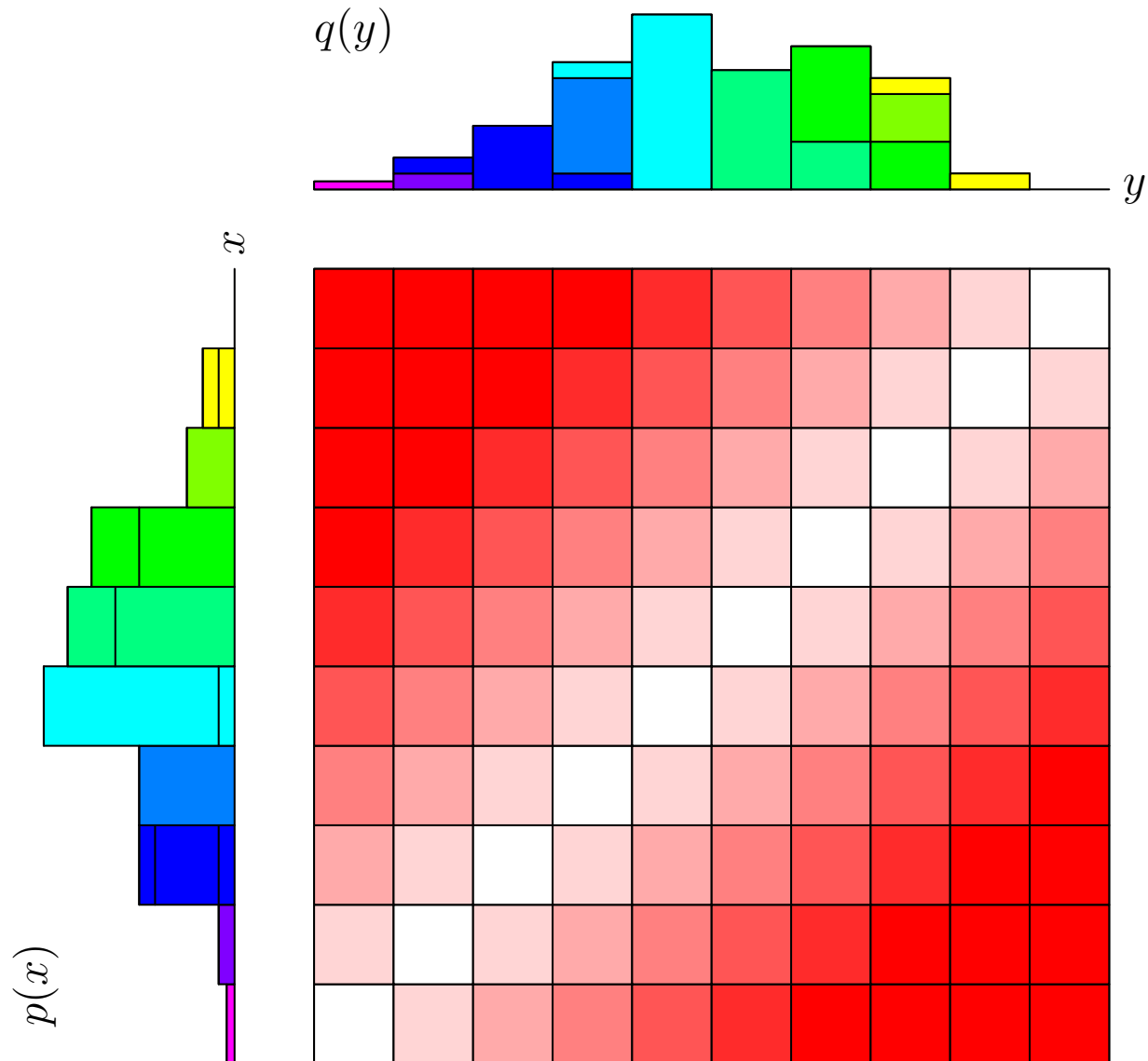
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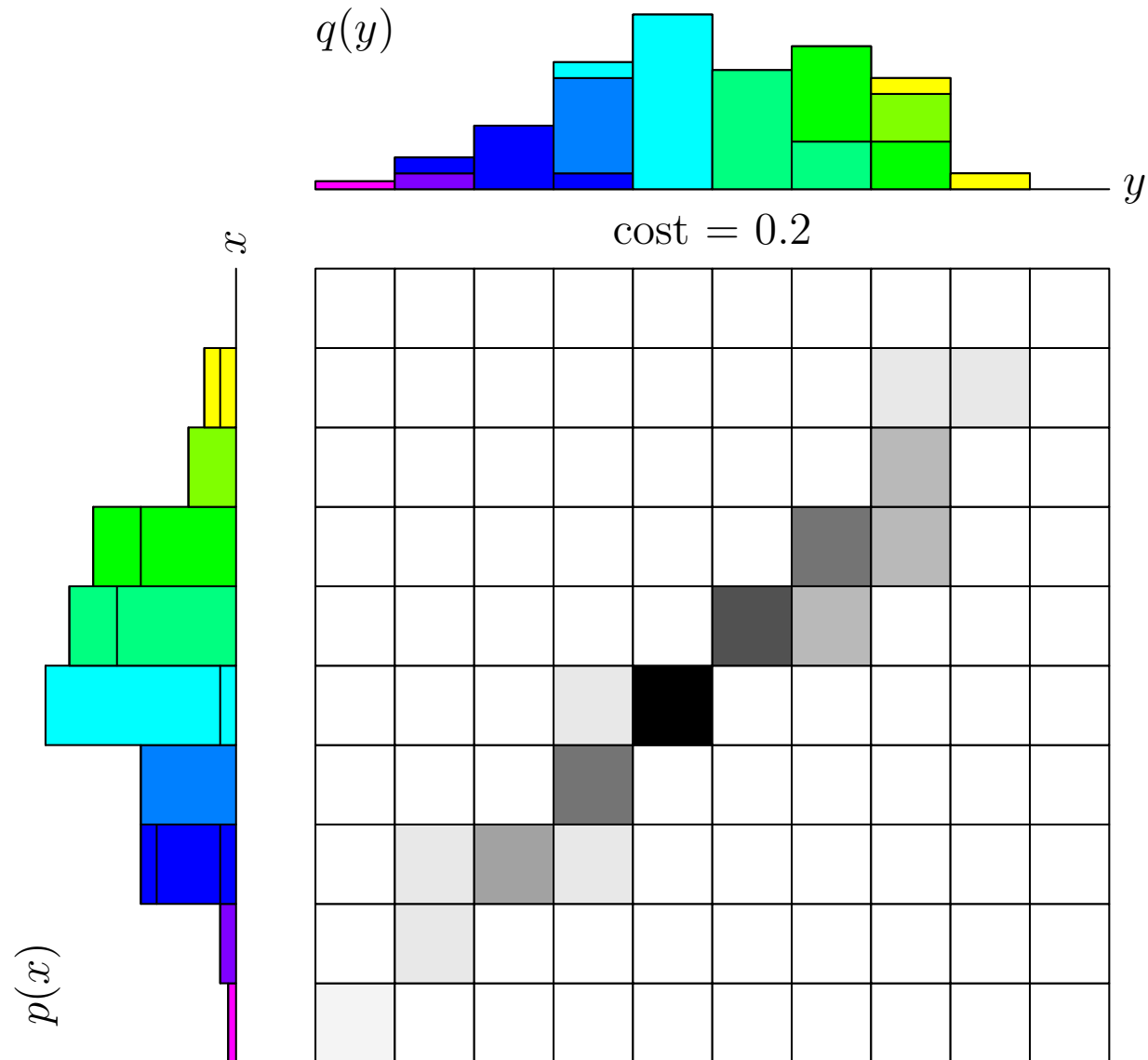
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The Wasserstein Distance

- The Wasserstein distance $W(p, q)$ between two probability distributions is defined as

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- Where $\Lambda(p, q)$ is the set of joint distributions $\gamma(\mathbf{x}, \mathbf{y})$ such that

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$$\int \gamma(\mathbf{x}, \mathbf{y}) d\mathbf{y} = p(\mathbf{x}) \qquad \int \gamma(\mathbf{x}, \mathbf{y}) d\mathbf{x} = q(\mathbf{y})$$

Computing the Wasserstein Distance

- To compute the Wasserstein distance we have to solve a minimisation task!
- This looks nasty, but it is a (continuous) linear programming problem
- Suppose p and q were discrete distribution (i.e. x and y only take discrete points)
- Then we could treat each value of $\gamma(x, y)$ as an element of a vector γ and each value of $d(x, y)$ as an element of a vector D
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Constraints

$$\sum_j \gamma(\mathbf{x}_i, \mathbf{y}_j) = p(\mathbf{x}_i)$$

$$\sum_i \gamma(\mathbf{x}_i, \mathbf{y}_j) = q(\mathbf{y}_j)$$

$$\mathbf{A} \boldsymbol{\gamma} = \mathbf{P}$$

$$\begin{pmatrix} 1 & 1 & \cdots & 1 & 0 & 0 & \cdots & 0 & \cdots & \cdots & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 1 & 1 & \cdots & 1 & \cdots & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \cdots & \cdots & 1 & 1 & \cdots & 1 \\ \hline 1 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 & \cdots & \cdots & 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & 0 & 1 & \cdots & 0 & \cdots & \cdots & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & 0 & \cdots & 1 & \cdots & \cdots & 0 & 0 & \cdots & 1 \end{pmatrix} \begin{pmatrix} \gamma(x_1, y_1) \\ \gamma(x_2, y_1) \\ \vdots \\ \gamma(x_n, y_1) \\ \hline \gamma(x_1, y_2) \\ \gamma(x_2, y_2) \\ \vdots \\ \gamma(x_n, y_2) \\ \hline \vdots \\ \vdots \\ \vdots \\ \hline \gamma(x_1, y_n) \\ \gamma(x_2, y_n) \\ \vdots \\ \gamma(x_n, y_n) \end{pmatrix} = \begin{pmatrix} q(y_1) \\ q(y_2) \\ \vdots \\ q(y_n) \\ \hline p(x_1) \\ p(x_2) \\ \vdots \\ p(x_n) \end{pmatrix}$$

Lagrange Formulation

- For discrete distributions

$$\begin{aligned} & \min_{\gamma} \mathbf{D}^T \gamma \\ & \text{subject to } \mathbf{A} \gamma = \mathbf{P}, \quad \gamma \geq 0 \end{aligned}$$

- Writing the Lagrangian

$$\mathcal{L}(\gamma, \alpha) = \mathbf{D}^T \gamma - \alpha^T (\mathbf{A}^T \gamma - \mathbf{P})$$

where α is a vector of Lagrange multipliers

- The solution to the discrete optimisation problem is given by

$$\min_{\gamma} \max_{\alpha} \mathcal{L}(\gamma, \alpha)$$

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Dual Form

- We can rearrange

$$\begin{aligned}\mathcal{L}(\gamma, \alpha) &= \mathbf{D}^\top \gamma - \alpha^\top (\mathbf{A} \gamma - \mathbf{P}) \\ &= \mathbf{P}^\top \alpha - \gamma^\top (\mathbf{A}^\top \alpha - \mathbf{D})\end{aligned}$$

- We note that $\gamma \geq 0$ so the dual problem is to find a vector α that maximises $\mathbf{P}^\top \alpha$ subject to the constraints $\mathbf{A}^\top \alpha \leq \mathbf{D}$
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Explicit Form

- We can write a Lagrangian for the original problem

$$\mathcal{L} = \sum_{i,j} d(\mathbf{x}_i, \mathbf{y}_j) \gamma(\mathbf{x}_i, \mathbf{y}_j) - \sum_i \alpha(\mathbf{x}_i) \left(\sum_j \gamma(\mathbf{x}_i, \mathbf{y}_j) - p(\mathbf{x}_i) \right) - \sum_j \beta(\mathbf{y}_j) \left(\sum_i \gamma(\mathbf{x}_i, \mathbf{y}_j) - q(\mathbf{y}_j) \right)$$

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- This is equivalent to maximising $\sum_i \alpha(\mathbf{x}_i) p(\mathbf{x}_i) + \sum_j \beta(\mathbf{y}_j) q(\mathbf{y}_j)$, subject to

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Dual Form Constraint

- We note that $\alpha(\mathbf{x}) + \beta(\mathbf{y}) \leq d(\mathbf{x}, \mathbf{y})$ for all $\mathbf{x}$ and $\mathbf{y}$
- This has to be true when $\mathbf{x} = \mathbf{y}$ so that

$$\alpha(\mathbf{x}) + \beta(\mathbf{x}) \leq d(\mathbf{x}, \mathbf{x}) = 0$$

- So $\beta(\mathbf{x}) = -\alpha(\mathbf{x}) - \epsilon(\mathbf{x})$ where $\epsilon(\mathbf{x}) \geq 0$
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Dual Form

- Thus the dual problem is to find a function $\alpha(\mathbf{x})$ —or a vector of functions $(\alpha(\mathbf{x}_i)|i)$ —that maximises

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- Subject to the constraint

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- This is a continuity constraint on the Lagrange multiplier function $\alpha(\mathbf{x})$ known as Lipschitz-1

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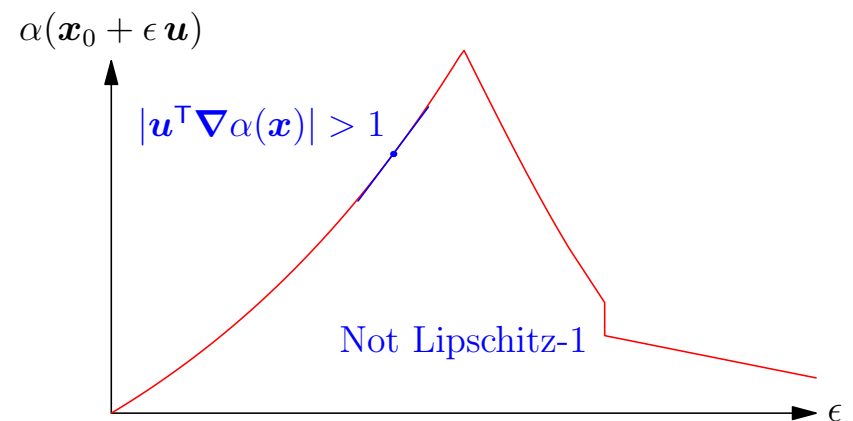
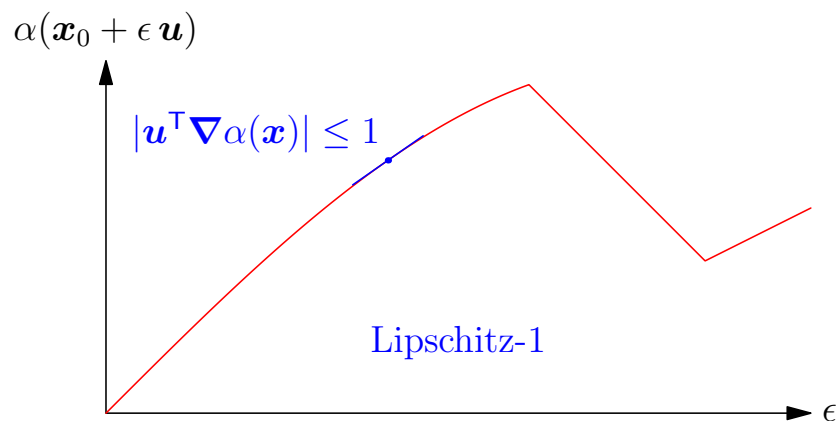
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Lipschitz-1 Functions

- We note for a Lipschitz-1 function and any unit vector $\mathbf{u}$

$$\mathbf{u}^\top \nabla \alpha(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{\alpha(\mathbf{x} + \epsilon \mathbf{u}) - \alpha(\mathbf{x})}{\epsilon}$$

- That is, at every point the gradient in all directions must be less than 1

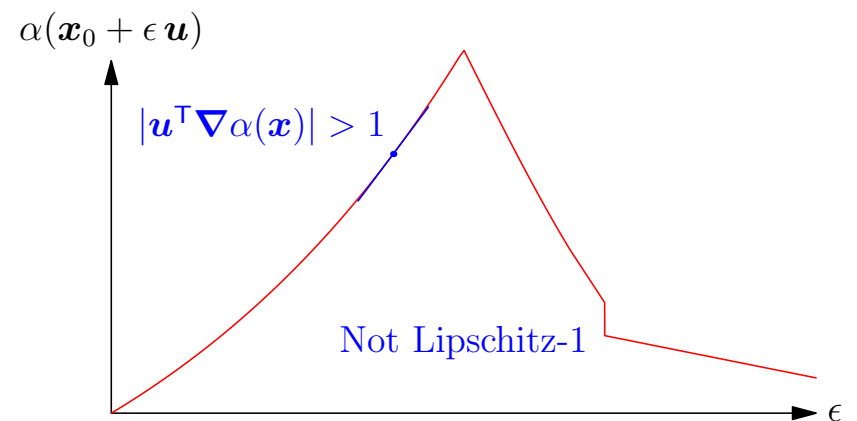
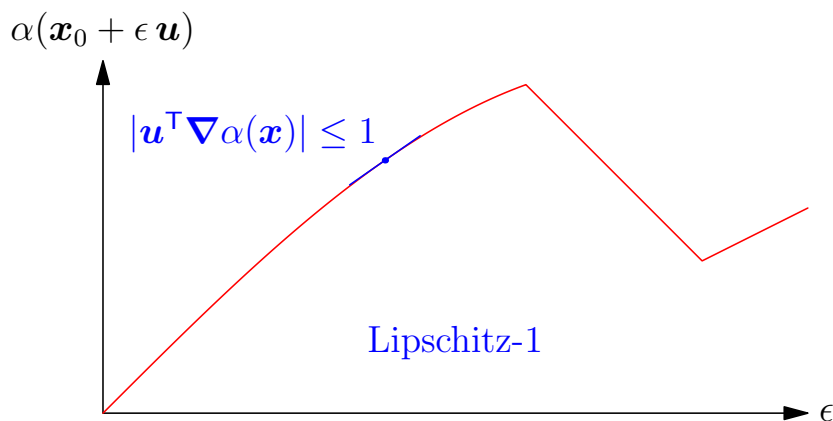


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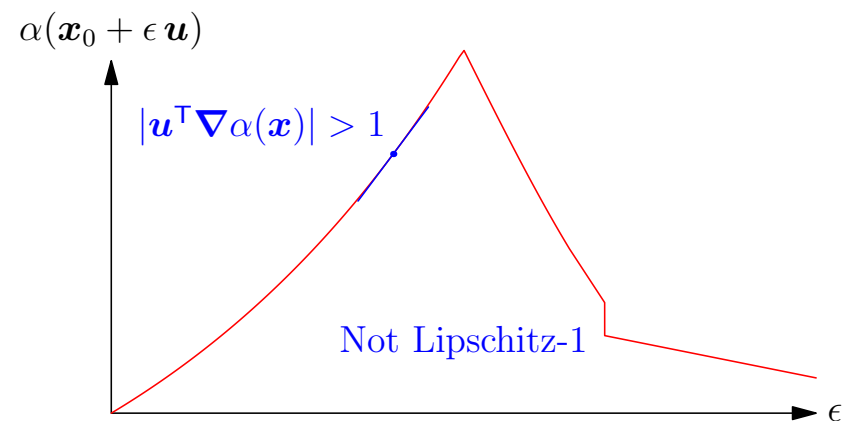
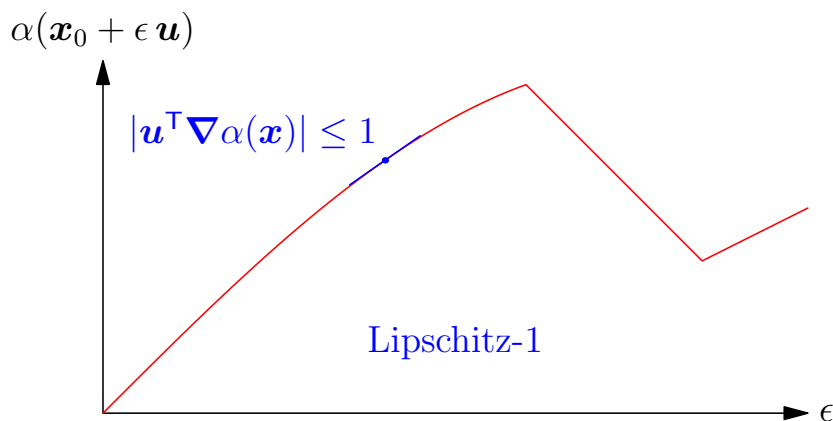


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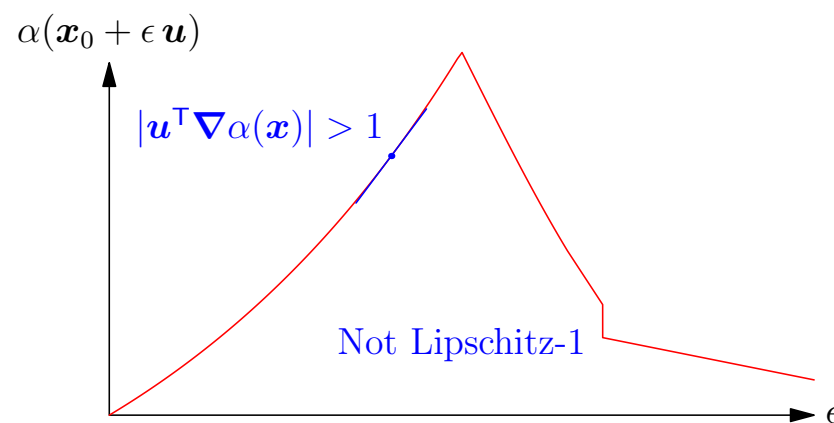
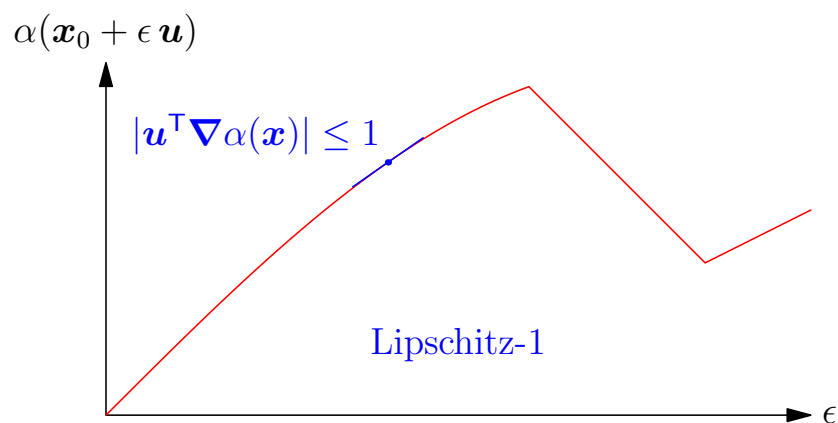


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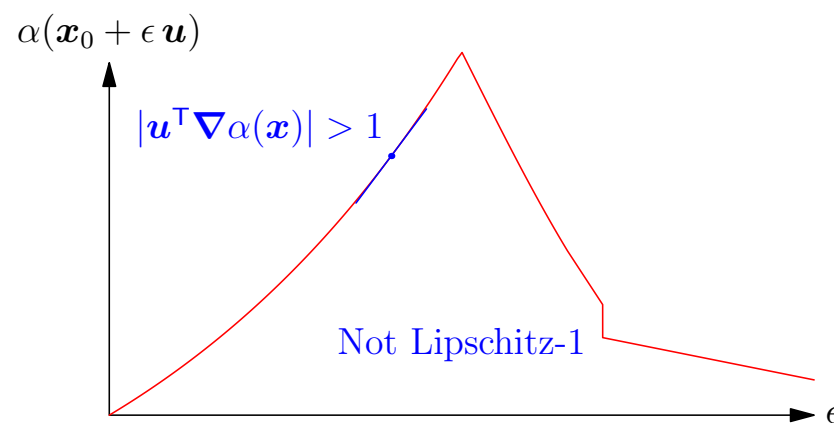
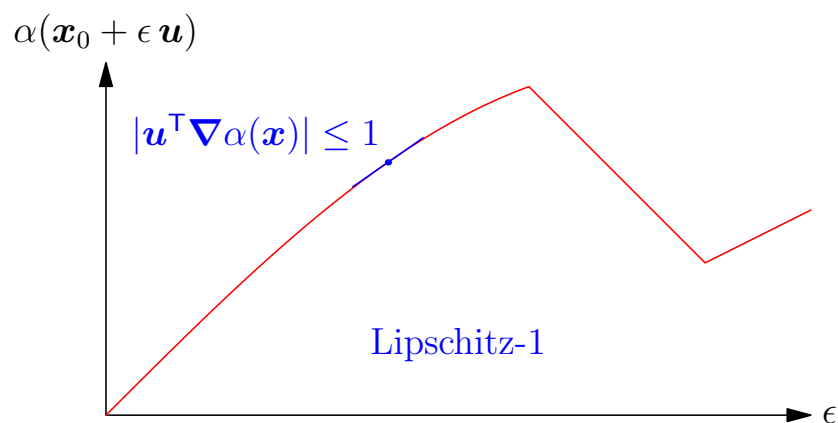


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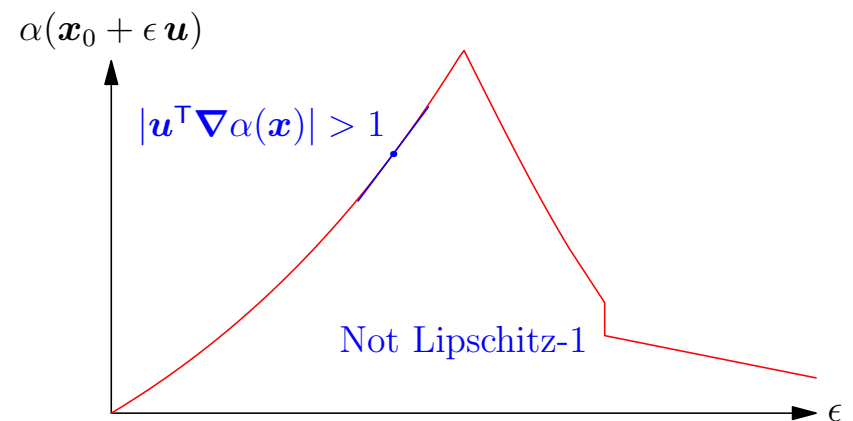
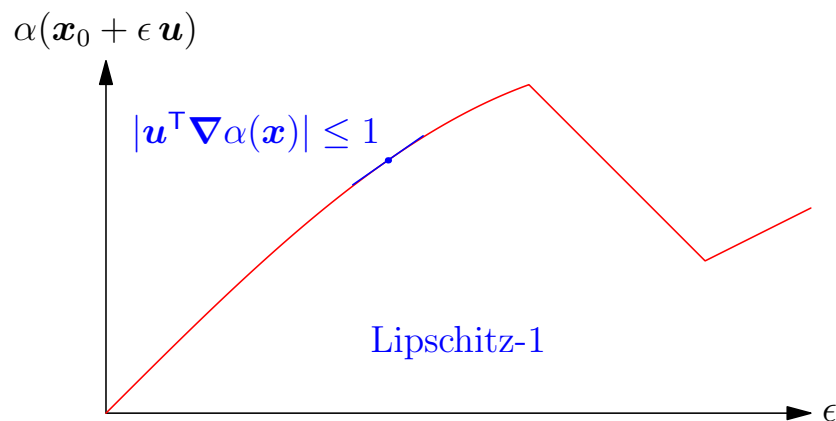


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Calculating the Wasserstein Distance

- To recall the big picture we want to compute the Wasserstein distance

$$W(p, q) = \min_{\gamma \in \Lambda(p, q)} \mathbb{E}_{\gamma}[d(\mathbf{x}, \mathbf{y})]$$

- For high dimensional objects $\gamma(\mathbf{x}, \mathbf{y})$ would be a huge object to approximate
- Instead we can compute the Wasserstein distance in the dual formulation

$$W(p, q) = \max_{\alpha(\mathbf{x})} \int \alpha(\mathbf{x}) (p(\mathbf{x}) - q(\mathbf{x})) d\mathbf{x}$$

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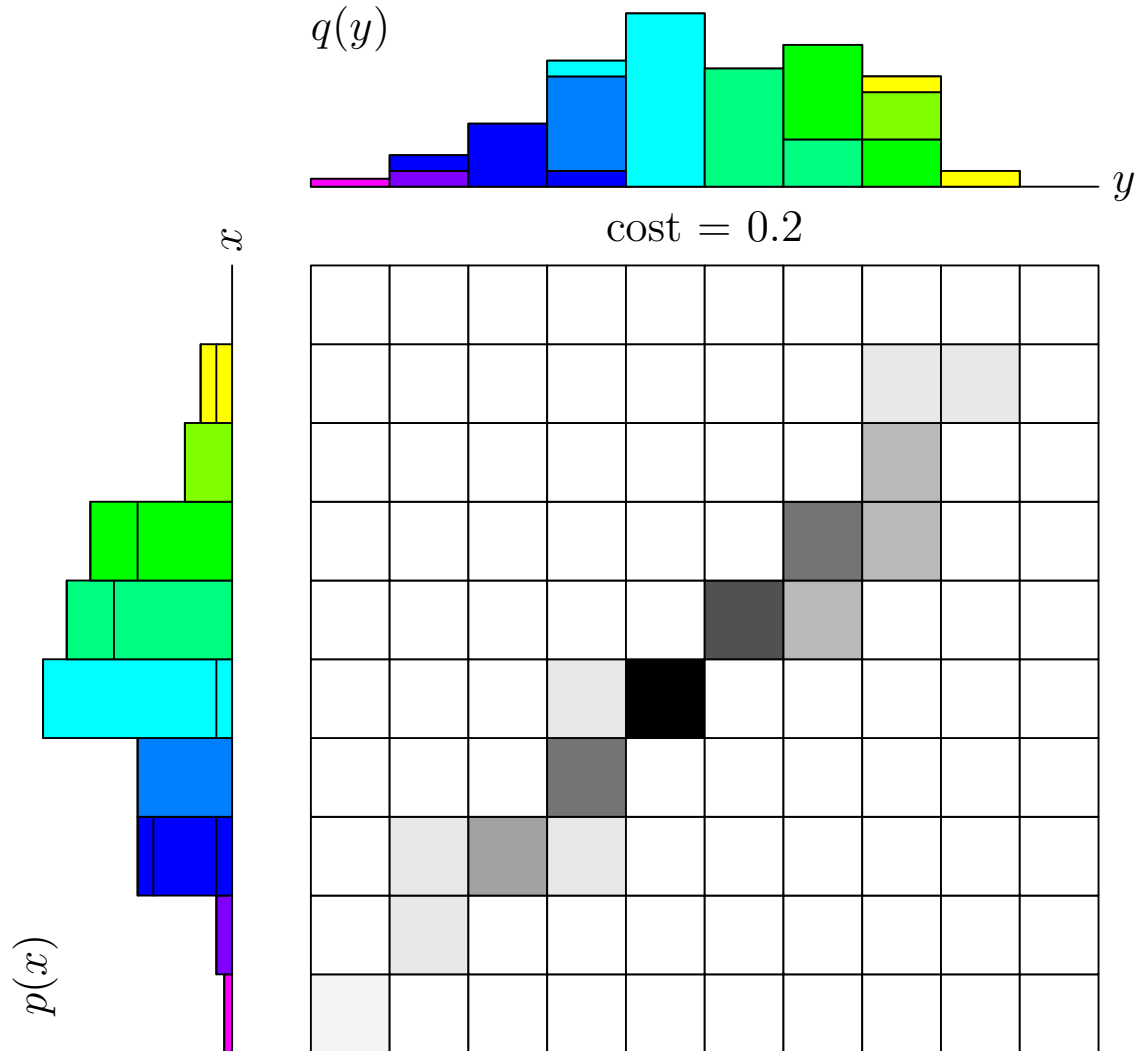
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Outline

1. GANs
2. Wasserstein Distance
3. **Wasserstein GANs**



Back to GANs

- What has this got to do with GANs?
- Suppose we want to minimise the distance between the distribution $p(\mathbf{x})$ of real images (of which $\mathcal{D}$ are samples) and the distribution $q(\mathbf{x})$ of images drawn from a generator
- We can use a normal GAN generator, $G(\mathbf{z}, \mathbf{w}_G)$, that generates an image when given a random variable $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- To do this we choose the weights, $\mathbf{w}_G$ of the generator to minimise

$$W(p, q) = \max_{\alpha(\mathbf{x})} (\mathbb{E}_{\mathbf{x} \sim p}[\alpha(\mathbf{x})] - \mathbb{E}_{\mathbf{x} \sim q}[\alpha(\mathbf{x})])$$

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- where $\mathcal{B} \subset \mathcal{D}$ is a minibatch of true images and $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
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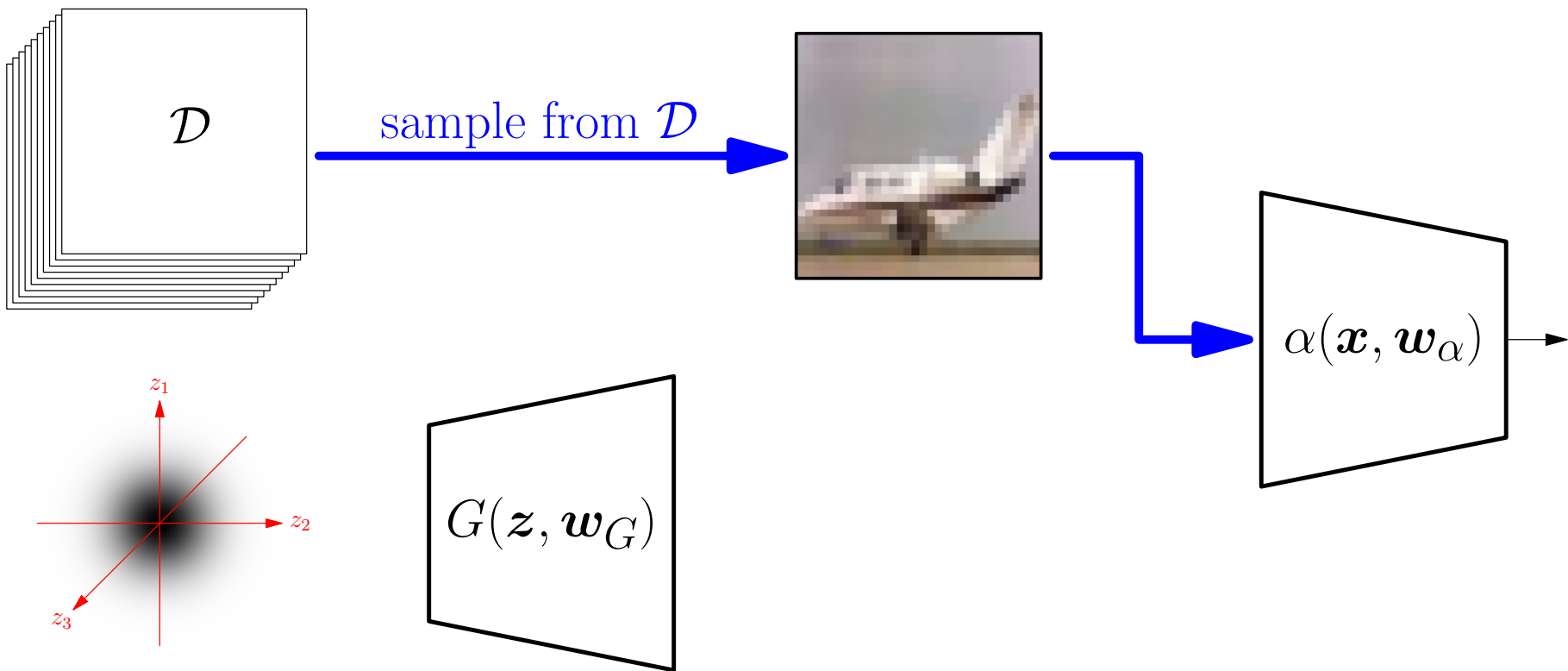
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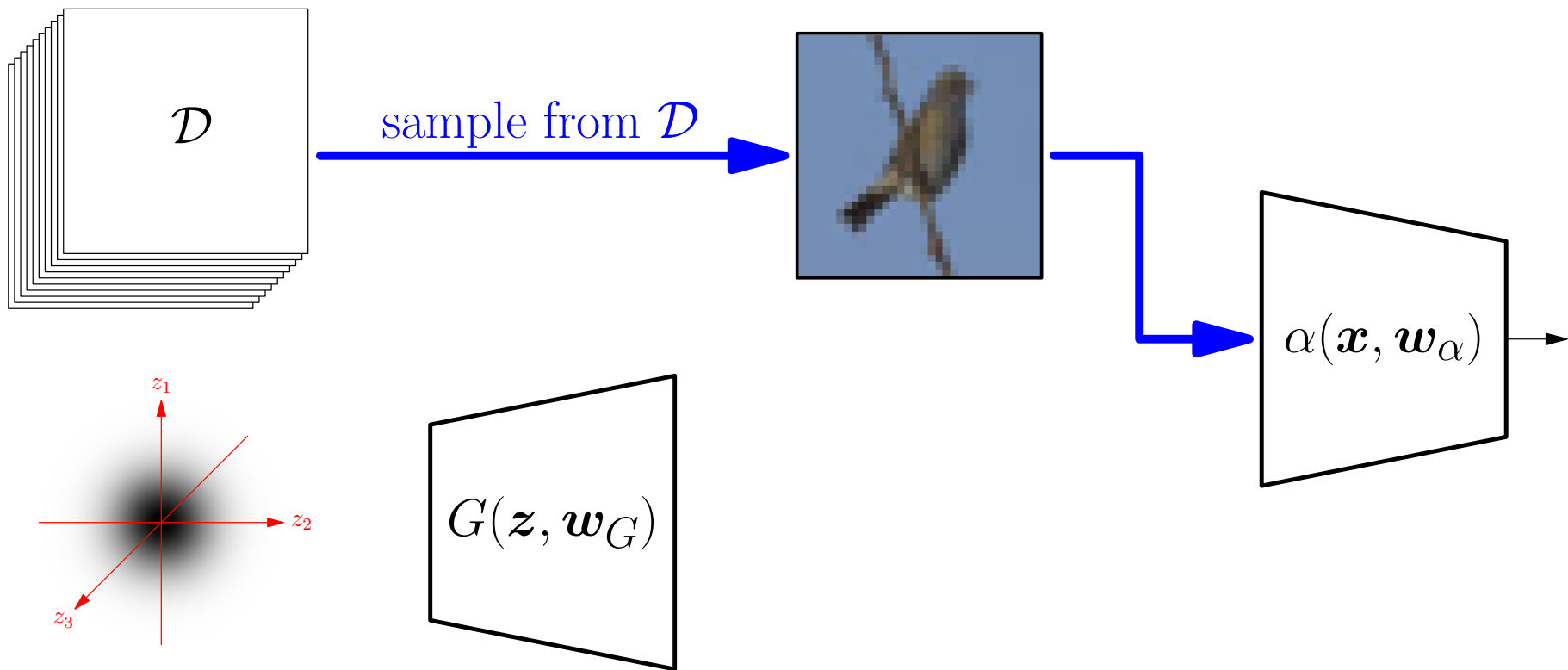
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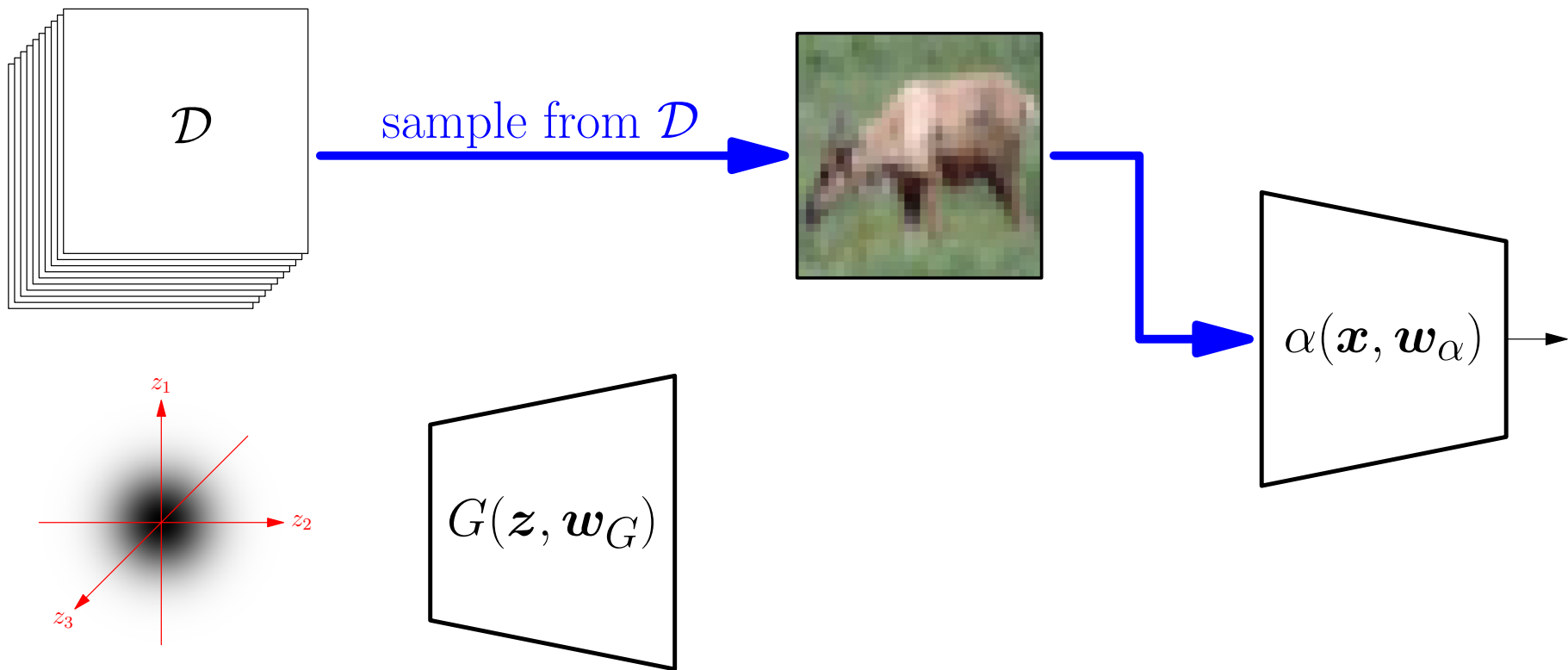
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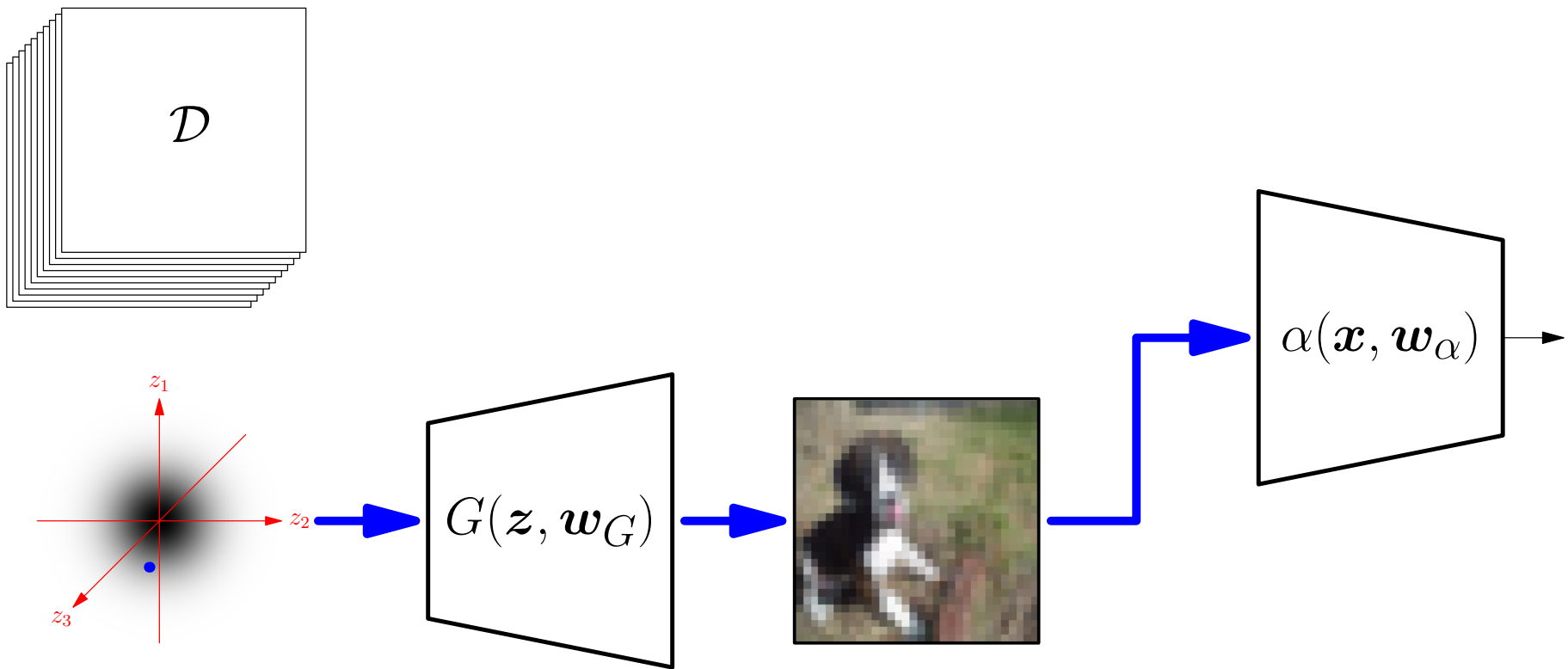
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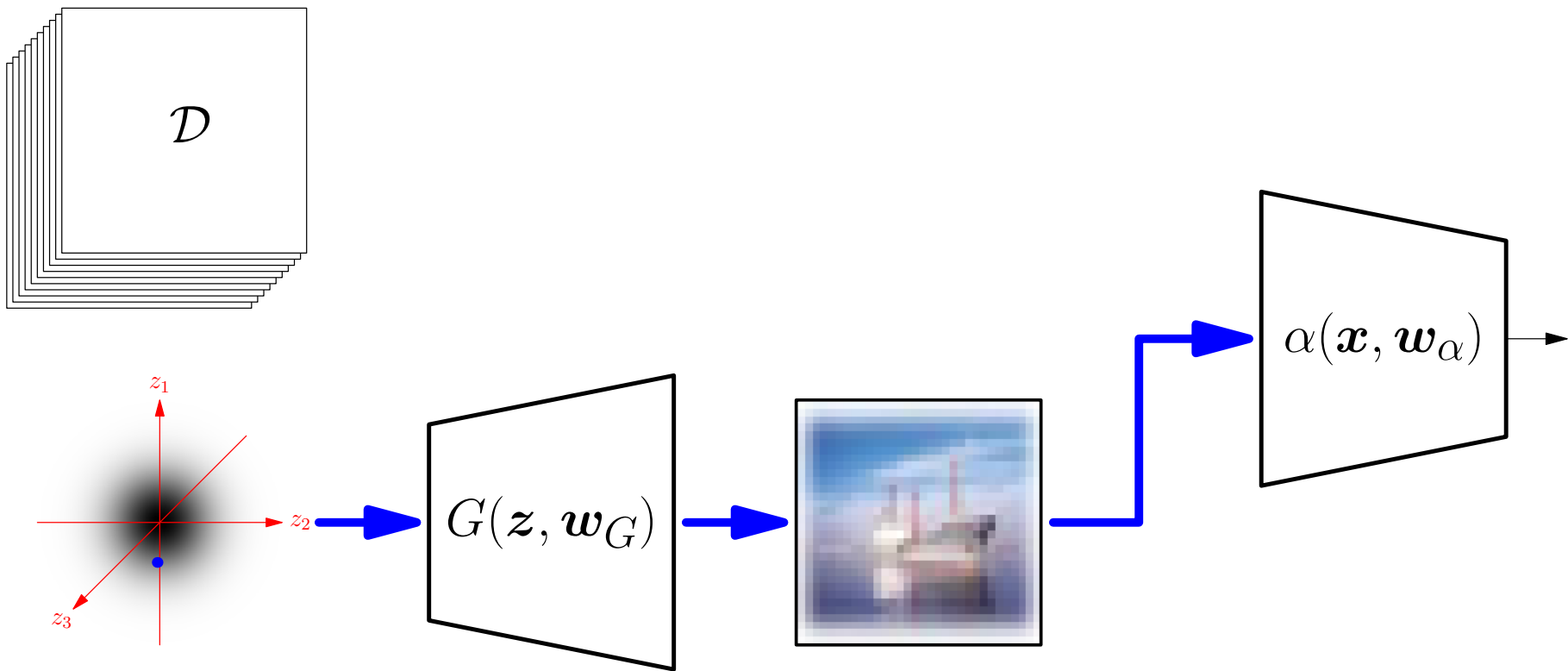
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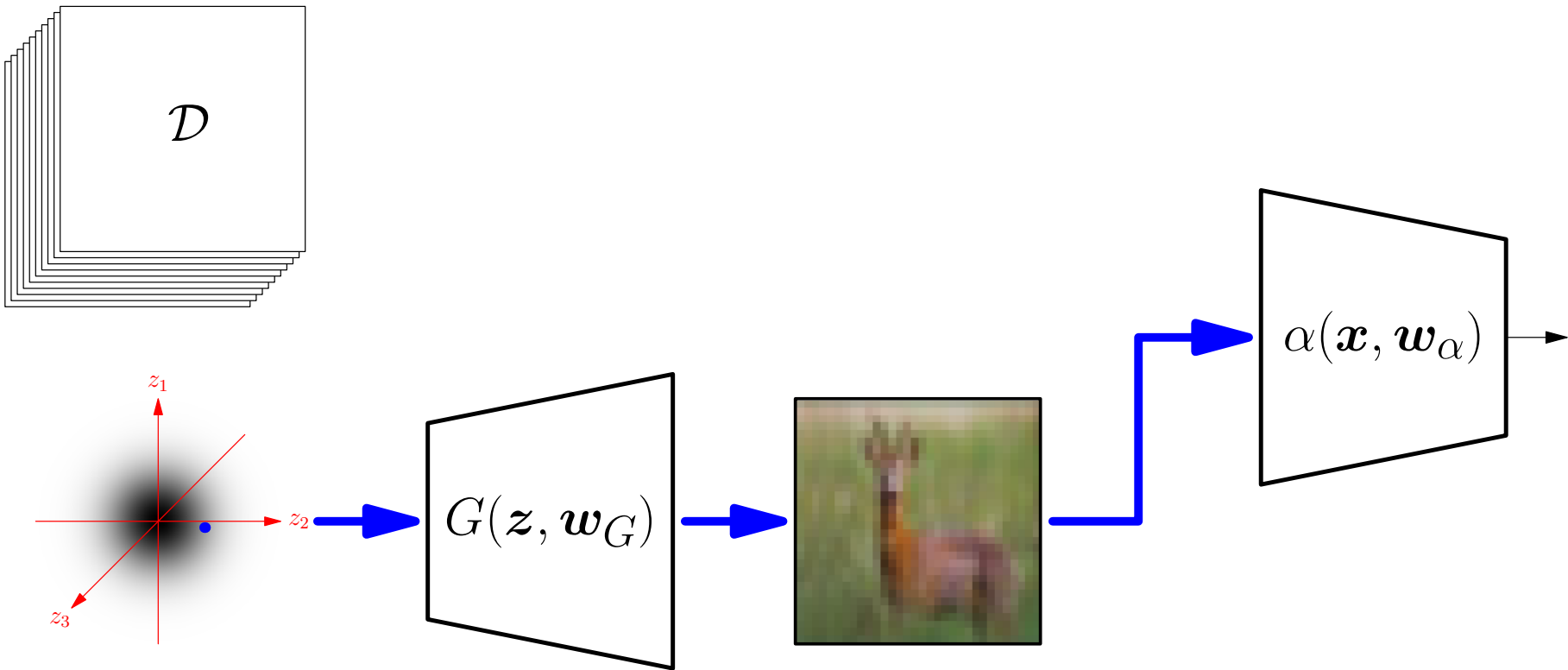
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- By trying to minimise the Wasserstein distance between the distribution of a generator and a true distribution we arrive at optimising two adversarial networks just like a GAN
- This uses a rather beautiful dual formulation
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